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# A role for arthropods as vectors of multidrug-resistant Enterobacterales in surgical site infections from South Asia

Brekhna Hassan[1] ✉, Muhammad Ijaz[2], Asadullah Khan[2], Kirsty Sands[1,3], Georgios-Ion Serfas[1], Liam Clayfield[1], Maisra Mohammed El-Bouseary[4], Giulia Lai[1], Edward Portal[1], Afifah Khan[5], William J. Watkins[1], Julian Parkhill[6] and Timothy R. Walsh[1,3]

**Understanding how multidrug-resistant Enterobacterales (MDRE) are transmitted in low- and middle-income countries (LMICs) is critical for implementing robust policies to curb the increasing burden of antimicrobial resistance (AMR). Here, we analysed samples from surgical site infections (SSIs), hospital surfaces (HSs) and arthropods (summer and winter 2016) to investigate the incidence and transmission of MDRE in a public hospital in Pakistan. We investigated Enterobacterales containing resistance genes (*bla*$_{CTX-M-15}$, *bla*$_{NDM}$ and *bla*$_{OXA-48}$-like) for identification, antimicrobial susceptibility testing and whole-genome sequencing. Genotypes, phylogenetic relationships and transmission events for isolates from different sources were investigated using single-nucleotide polymorphism (SNP) analysis with a cut-off of ≤20 SNPs. *Escherichia coli* (14.3%), *Klebsiella pneumoniae* (10.9%) and *Enterobacter cloacae* (16.3%) were the main MDRE species isolated. The carbapenemase gene *bla*$_{NDM}$ was most commonly detected, with 15.5%, 15.1% and 13.3% of samples positive in SSIs, HSs and arthropods, respectively. SNP (≤20) and spatiotemporal analysis revealed linkages in bacteria between SSIs, HSs and arthropods supporting the One Health approach to underpin infection control policies across LMICs and control AMR.**

AMR is a global problem that is being exacerbated by the rapid increase in infections by MDRE and extensively drug-resistant Enterobacterales (XDRE)[1]. Most reports on the dissemination of AMR are primarily patient-centric and hospital-based[2]. However, One Health epidemiological studies have found that MDRE/XDRE are widely disseminated in the environment and are linked to human infections[3]. Extended-spectrum β-lactamase (ESBLs) as well as carbapenemase-positive bacteria that have been isolated from the environment, animal and food sources have substantial genetic identity to pathogens that cause human infections; however, studies definitively proving connectivity between environmental organisms and human infection are limited[3,4].

South Asian countries such as India, Bangladesh and Pakistan have a high incidence of MDRE/XDRE, due in part to the overuse and misuse of antibiotics in human medicine and agriculture[5,6]. However, in these LMICs, there are additional factors to consider with regard to AMR, including flooding, substandard sanitation and environmental contamination (including hospital waste management)[7,8]. In addition to human carriage, transmission of MDRE between the outside environment and hospitals in LMICs might also be propagated by arthropods, particularly by flies that feed on organic decaying matter, including human and animal faeces. Recently, there have been several reports examining the transmission of MDRE by flies, including studies from China, Brazil, Japan, United States, Germany, Spain and Thailand; however, these studies were limited to farms and their immediate communities[9–15]. Moreover, there are only a few studies examining other insects, for example, cockroaches and carriage of MDRE, and these are mainly derived from LMICs, including North Africa/Persia, sub-Sahara Africa and South Asia[16–18]. Surprisingly, there is a lack of studies linking MDRE dissemination to spiders and, although the aforementioned studies are informative, they lack a rigorous sampling strategy with whole-genome sequencing (WGS) to obtain evidence for circulating clades of bacteria within a specified time line between arthropods (insects and spiders), HSs and SSIs.

In this Article, we combine patient demographics, comprehensive sampling in two seasons (comparing summer to winter) and in-depth genomics to analyse the connectivity between arthropods, HSs and SSIs at a tertiary care hospital in Peshawar, Pakistan.

## Results

**Sampling strategy.** Samples were collected from the Khyber Teaching Hospital (KTH)) in Peshawar, Pakistan and consisted of arthropods and HS and SSI swabs. Arthropod samples consisted of a single insect or spider. The study outline is described in Fig. 1 with the location illustrated in Supplementary Figs. 1 and 2. A two-point surveillance system was established by collecting in peak winter (January 2016) $n=152$, $n=316$ and $n=726$ SSI, hospital surface and arthropod samples, respectively, and in summer (July–August 2016) $n=190$, $n=308$ and $n=1,253$ SSI, hospital surface and arthropod samples, respectively. Insects captured at the time of sampling were identified as *Musca domestica*

[1]Institute of Infection and Immunity, Cardiff University, Cardiff, UK. [2]Khyber Teaching Hospital, Peshawar, Pakistan. [3]Ineos Institute of Antimicrobial Research, Department of Zoology, University of Oxford, Oxford, UK. [4]Pharmaceutical Microbiology Department, Faculty of Pharmacy, Tanta University, Tanta, Egypt. [5]School of Biosciences, Cardiff University, Cardiff, UK. [6]Department of Veterinary Medicine, University of Cambridge, Cambridge, USA. ✉e-mail: brekhnahassan@hotmail.com

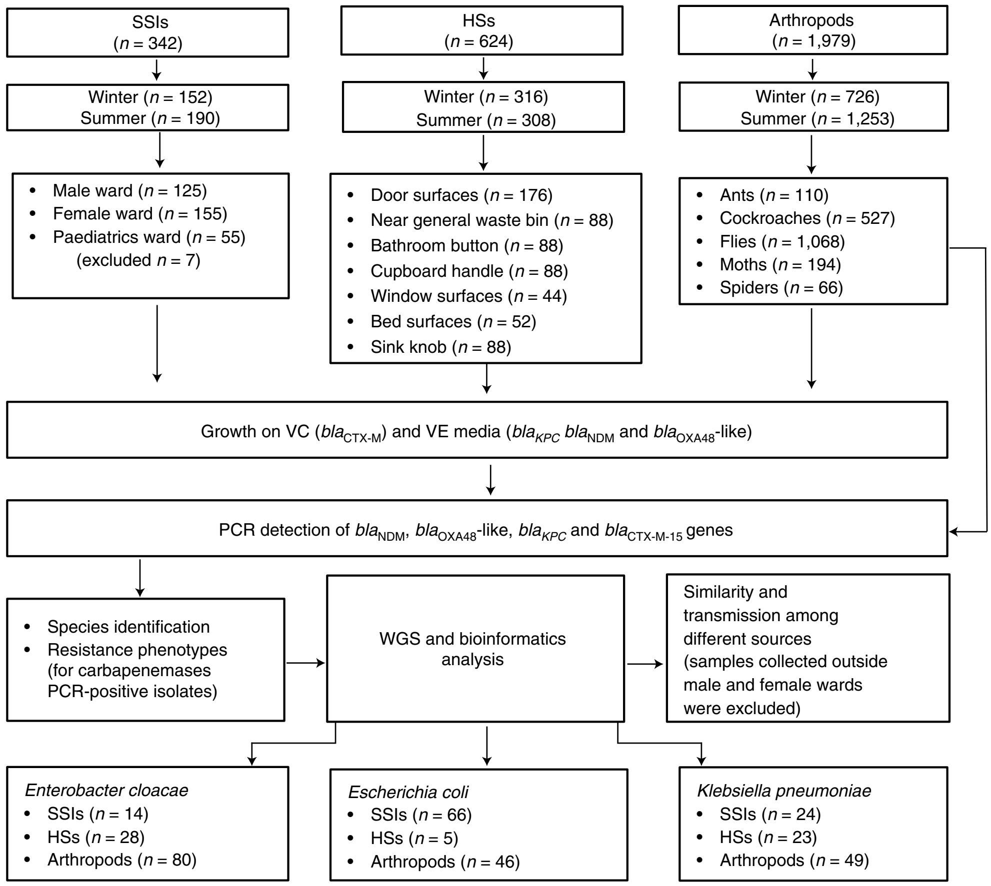



**Fig. 1 | Study flowchart.** The number of SSI, HS and arthropod samples collected in both seasons is shown, including the wards for SSIs, HSs and arthropods, the exact locations of the sampling sites for HS samples, and the species of arthropods for arthropods samples. The study outline is shown detailing the methodology used to process the samples. Any samples collected from outside the male and female wards were excluded from the SNP transmission analysis.

(house flies), *Blattella orientalis* or *Blattella germanica* (cockroaches) (Supplementary Fig. 3), *Lasius niger* (ants) and *Aglossa aglossalis* (moths). Spiders were identified as *Stegodyphus pacificus*. Both *B. orientalis* and *B. germanica* are common in South Asia[19,20]. Spiders, moths, ants and cockroaches were mostly caught alive whereas a fly trap was used to capture the flies. Before the samples were taken, routine cleaning procedures of the hospital were performed. Most SSIs (83.92%, 287/342) were collected from adult male and female wards; however, $n=55$ SSI samples were also collected from paediatric units (Fig. 1). SSIs are deemed to be post-surgical wounds presenting with post-surgery fever, wound dehiscence or wound site puss, exudate or odour and align with CDC SSI criteria[21] (Supplementary Table 1). The collection of hospital surface and arthropod samples was restricted to only male and female wards. Furthermore, all hospital surface sampling sites were kept constant and the same locations were sampled in both seasons, including door surfaces, near non-clinical waste, washroom button, cupboard handle, bed surfaces window surfaces and sinks. Arthropods were collected from inside or within the immediate vicinity of these wards. Most patients (79.2%, 271/342) were administered β-lactam prophylactic coverage; extended-spectrum cephalosporins (56.1%, 192/342) were the most popular and were regularly used in combination with sulbactam (30.1%, 103/342). Thus, genes conferring resistance to β-lactams such as $bla_{\text{CTX-M}}$ (global exemplar of ESBL) and the carbapenemase genes $bla_{\text{NDM}}$, $bla_{\text{OXA48}}$-like and $bla_{\text{KPC}}$ were selected as target resistant mechanisms for this study.

**The prevalence of $bla_{\text{CTX-M}}$ and carbapenemase genes.** All samples were cultured on vancomycin ($10\,\mathrm{mg\,l^{-1}}$) and cefotaxime ($1\,\mathrm{mg\,l^{-1}}$) (VC), and vancomycin ($10\,\mathrm{mg\,l^{-1}}$) and ertapenem ($1\,\mathrm{mg\,l^{-1}}$) (VE) selective media to select for Gram-negative bacteria resistant to cefotaxime and ertapenem, respectively. All samples from SSIs and HSs displayed growth on both VC and VE media; however, from arthropods, 88.8% and 88.4% of samples grew on VC and VE, respectively. $bla_{\text{CTX-M-15}}$ was highest among all sources (for SSIs, HSs and arthropods, 29.8% (102/342), 21.8% (136/624) and 35.4% (700/1,979), respectively) followed by $bla_{\text{NDM}}$ (for SSIs, HSs and arthropods, 15.5% (53/342), 13.3% (83/624) and 15.2% (300/1,979), respectively) and $bla_{\text{OXA-48}}$-like (for SSIs, HSs and arthropods, 13.5% (46/342), 3.2% (20/624) and 2.6% (51/1,979), respectively) (Extended Data Fig. 1). $bla_{\text{KPC}}$ was not detected in any of our samples.

Among the different arthropod species, $bla_{\text{NDM}}$ was observed in 27.0% (144/533) of cockroaches, 5.2% (10/66) of spiders, 11.6% (125/1,076) of flies, 9.3% (18/194) of moths and 2.7% (3/110) of ants. $bla_{\text{CTX-M-15}}$ was observed in 45.0% (240/533) of cockroaches, 36.4% (24/66) of spiders, 34.6% (372/1,076) of flies, 28.4% (55/194) of moths and 8.2% (9/110) of ants. $bla_{\text{OXA-48}}$-like was observed in 5.8% (31/533) of cockroaches, 2.7% (3/110) of ants, 1.5% (3/194) of moths, 1.5% (1/66) of spiders and 1.2% (13/1,076) of flies (Fig. 2 and Supplementary Table 2).

**Seasonal variation of MDRE in samples.** To understand the seasonal variations between samples, we compared the carriage of

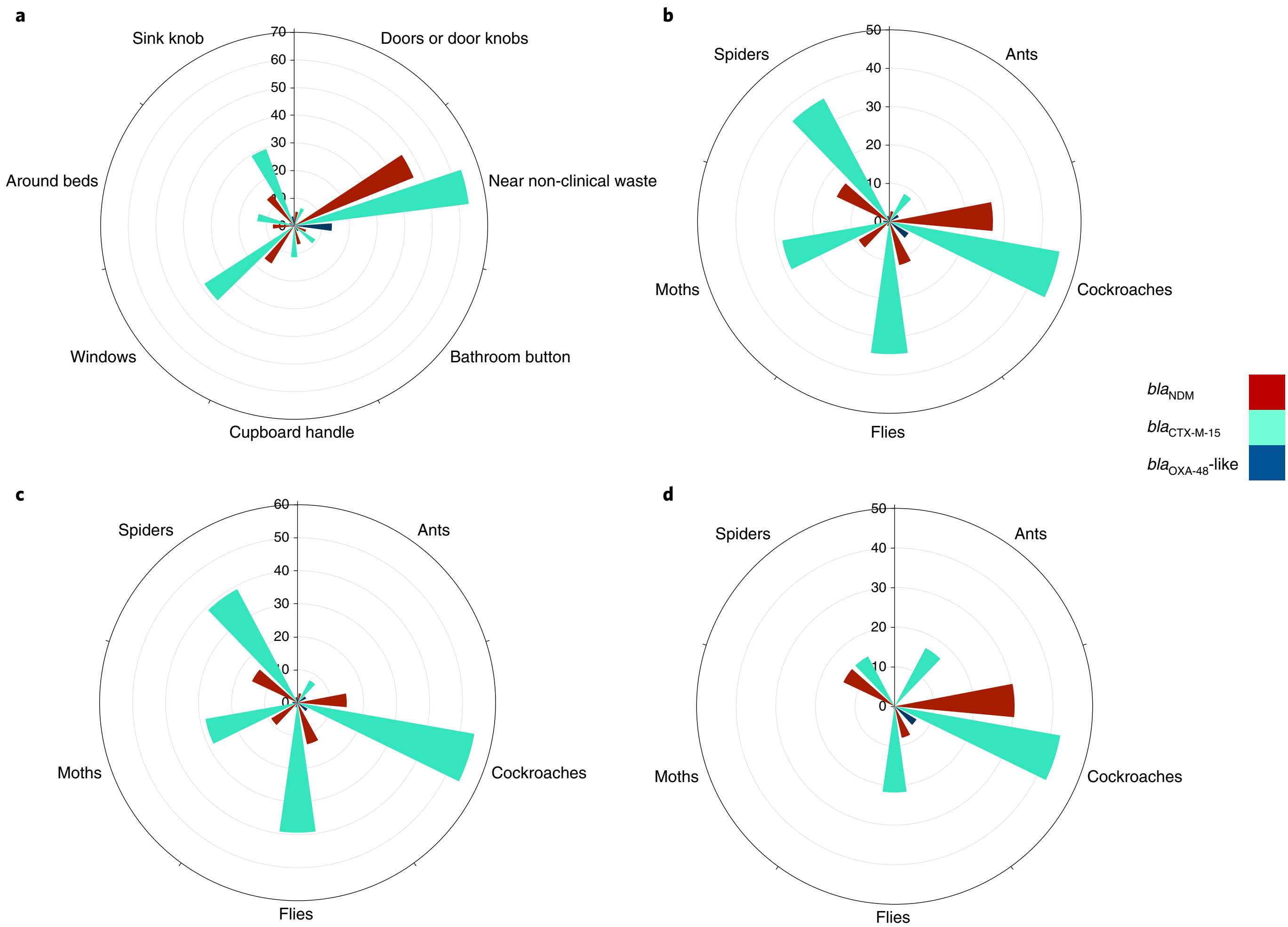



**Fig. 2 | The distribution of resistance genes among HSs and species of arthropods represented as rose graphs. a**, The distribution of $bla_{NDM}$, $bla_{OXA-48}$-like and $bla_{CTX-M-15}$ among different HS sites collected in summer and winter. **b**, The distribution of $bla_{NDM}$, $bla_{OXA-48}$-like and $bla_{CTX-M-15}$ among different species of arthropods collected in summer and winter. **c**, The distribution of $bla_{NDM}$, $bla_{OXA-48}$-like and $bla_{CTX-M-15}$ among different species of arthropods collected in summer. **d**, The distribution of $bla_{NDM}$, $bla_{OXA-48}$-like and $bla_{CTX-M-15}$ among different species of arthropods collected in winter. Data are percentages and the target genes are represented by different colours.

$bla_{NDM}$, $bla_{OXA-48}$-like and $bla_{CTX-M-15}$ across different seasons. Among SSIs and arthropods, a higher frequency of carbapenemase genes was observed in the winter (SSIs, for $bla_{NDM}$, $bla_{OXA-48}$-like and $bla_{CTX-M-15}$, 21.1% (32/152), 19.7% (30/152) and 40.1% (61/152), respectively; arthropods, for $bla_{NDM}$, $bla_{OXA-48}$-like and $bla_{CTX-M-15}$, 20.9% (152/726), 4.0% (29/726) and 33.6% (244/726), respectively) compared to summer (SSIs, for $bla_{NDM}$, $bla_{OXA-48}$-like and $bla_{CTX-M-15}$, 11.1% (21/190), 8.4% (16/190) and 21.6% (41/190), respectively; arthropods, for $bla_{NDM}$, $bla_{OXA-48}$-like and $bla_{CTX-M-15}$, 11.8% (148/1,253), 1.8% (22/1,253) and 36.4% (456/1,253), respectively). By contrast, a marginally higher prevalence was displayed from HS samples in the winter (for $bla_{NDM}$, $bla_{OXA-48}$-like and $bla_{CTX-M-15}$, 15.5% (49/316), 3.8% (12/316) and 25.6% (81/316), respectively) compared to summer (for $bla_{NDM}$, $bla_{OXA-48}$-like and $bla_{CTX-M-15}$, 11.0% (34/308), 2.6% (8/308) and 17.9% (55/308), respectively) (Extended Data Fig. 1). The observed relationship between seasons and carriage of resistance genes was statistically significant for $bla_{NDM}$ (odds ratio (OR) = 3.5, 95% confidence interval (CI) = 1.6–7.5), $bla_{OXA-48}$-like (OR = 2.4, 95% CI = 1.2–4.8) and $bla_{CTX-M-15}$ (OR = 2.5, 95% CI = 1.4–4.4) among SSIs; and $bla_{NDM}$ (OR = 2.0, 95% CI = 1.5–2.5) and $bla_{OXA-48}$-like (OR = 2.3, 95% CI = 1.3–4.0) among arthropods when compared with the baseline category of season summer (Table 1). Multivariate analysis among significant variables was concordant (Supplementary Table 3).

The majority of our arthropod samples consisted of flies (*M. domestica*, 54.0%) and cockroaches (26.6%). The local climate is dry with average temperatures for summer and winter of 33 °C and 14 °C, respectively. A warmer climate favours the prevalence of flies and other insects, which is consistent with the increased abundance collected in our study in summer beside cockroaches, of which the majority (78.6%, 419/533) were collected in the winter. Flies and cockroaches showed the highest prevalence of target resistant genes in both seasons (Fig. 2 and Supplementary Table 2); however, we observed an increased percentage of MDRE isolated from all arthropod species in the summer.

**MDRE and arthropod species.** We examined the risk factors (Table 1) for MDRE cultured from SSIs and found significant associations with longer hospital stays of >4–6 d ($bla_{NDM}$, OR = 3.6, 95% CI = 1.3–10.0; $bla_{OXA-48}$-like, OR = 3.7, 95% CI = 1.4–9.4; and $bla_{CTX-M-15}$, OR = 2.3, 95% CI = 1.2–4.5) and ≥7 d ($bla_{NDM}$, OR = 9.9, 95% CI = 3.6–27.1; $bla_{OXA-48}$-like, OR = 8.4, 95% CI = 3.2–22.0; and $bla_{CTX-M-15}$, OR = 3.6, 95% CI = 1.7–7.3) when compared with the baseline of 1–3 d hospital stay. We also found a significant association

**Table 1 | Associations of *bla*$_{NDM}$, *bla*$_{OXA-48}$-like and *bla*$_{CTX-M-15}$ among SSIs, HSs and arthropods**

| Source | Variables | | Positive (*n*) | Positive (%) | Total (*n*) | Overall significance | Significance (compared to the reference category[a]) | OR | 95% CI for OR | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | Lower | Upper |
| SSIs | *bla*$_{NDM}$ | | | | | | | | | |
| | Season | Summer[a] | 11 | 7.48 | 147 | NA | 0.001 | 3.5 | 1.6 | |
| | | **Winter** | 25 | 22.12 | 113 | | | | | |
| | Stay | 1–3 d[a] | 6 | 4.80 | 125 | <0.001 | 0.012 | 3.6 | 1.3 | 10.0 |
| | | **4–6 d** | 13 | 15.48 | 84 | | <0.001 | 9.9 | 3.6 | 27.1 |
| | | **≥7 d** | 17 | 33.33 | 51 | | | | | |
| | Age | 14–30[a] | 13 | 15.66 | 83 | | | | | |
| | | 31–45 | 8 | 10.81 | 74 | 0.813 | 0.38 | 0.7 | 0.3 | 1.7 |
| | | 46–60 | 11 | 13.92 | 79 | | 0.76 | 0.9 | 0.4 | 2.1 |
| | | >61 | 4 | 16.67 | 24 | | 0.91 | 1.1 | 0.3 | 3.7 |
| | Ward | Male[a] | 22 | 18.33 | 120 | | | | | |
| | | Female | 24 | 10.00 | 140 | | 0.06 | 0.5 | 0.2 | 1.0 |
| | *bla*$_{CTX-M-15}$ | | | | | | | | | |
| | Season | Summer[a] | 29 | 19.73 | 147 | NA | 0.001 | 2.5 | 1.4 | 4.4 |
| | | **Winter** | 43 | 38.05 | 113 | | | | | |
| | Stay | 1–3 d[a] | 22 | 17.60 | 125 | 0.001 | 0.010 | 2.3 | 1.2 | 4.5 |
| | | **4–6 d** | 28 | 33.33 | 84 | | 0.001 | 3.6 | 1.7 | 7.3 |
| | | **≥7 d** | 22 | 43.14 | 51 | | | | | |
| | Age | 14–30[a] | 17 | 20.48 | 83 | 0.353 | 0.18 | 1.6 | 0.8 | 3.4 |
| | | 31–45 | 22 | 29.73 | 74 | | 0.11 | 1.8 | 0.9 | 3.7 |
| | | 46–60 | 25 | 31.65 | 79 | | 0.20 | 1.9 | 0.7 | 5.3 |
| | | >61 | 8 | 33.33 | 24 | | | | | |
| | Ward | Male[a] | 38 | 31.67 | 120 | NA | 0.19 | 0.7 | 0.4 | 1.2 |
| | | Female | 34 | 24.29 | 140 | | | | | |
| | *bla*$_{OXA-48}$-like | | | | | | | | | |
| | Season | Summer[a] | 15 | 10.20 | 147 | NA | | | | |
| | | **Winter** | 24 | 21.24 | 113 | | 0.015 | 2.4 | 1.2 | 4.8 |
| | Stay | 1–3 d[a] | 7 | 5.60 | 125 | <0.001 | 0.007 | 3.7 | 1.4 | 9.4 |
| | | **4–6 d** | 15 | 17.86 | 84 | | <0.001 | 8.4 | 3.2 | 22.0 |
| | | **≥7 d** | 17 | 33.33 | 51 | | | | | |
| | Age | 14–30[a] | 14 | 16.87 | 83 | 0.586 | 0.41 | 0.7 | 0.3 | 1.7 |
| | | 31–45 | 9 | 12.16 | 74 | | 0.89 | 1.1 | 0.5 | 2.4 |
| | | 46–60 | 14 | 17.72 | 79 | | 0.31 | 0.4 | 0.1 | 2.1 |
| | | >61 | 2 | 8.33 | 24 | | | | | |
| | Ward | Male[a] | 21 | 17.50 | 120 | NA | | | | |
| | | Female | 18 | 12.86 | 140 | | 0.30 | 0.7 | 0.4 | 1.4 |
| Arthropods | *bla*$_{NDM}$ | | | | | | | | | |
| | Arthropods | Moths[a] | 18 | 9.28 | 194 | | | | | |
| | | Ants | 3 | 2.73 | 110 | | 0.042 | 0.3 | 0.1 | 1.0 |
| | | Spiders | 10 | 15.15 | 66 | | 0.19 | 1.7 | 0.8 | 4.0 |
| | | **Cockroaches** | 144 | 27.32 | 527 | | <0.001 | 3.7 | 2.2 | 6.2 |
| | | Flies | 125 | 11.70 | 1,068 | | 0.33 | 1.3 | 0.8 | 2.2 |
| | Season | Summer[a] | 148 | 11.95 | 1,239 | NA | <0.001 | 2.0 | 1.5 | 2.5 |
| | | Winter | 152 | 20.94 | 726 | | | | | |
| | Ward | Male[a] | 94 | 14.78 | 636 | NA | 0.68 | 1.1 | 0.8 | 1.4 |
| | | **Female** | 206 | 15.50 | 1,329 | | | | | |

Continued

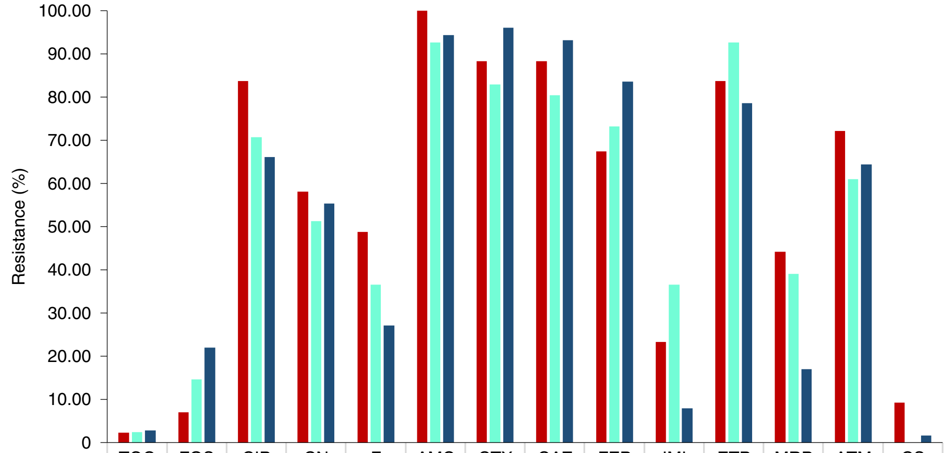
**Table 1 | Associations of $bla_{NDM}$, $bla_{OXA-48}$-like and $bla_{CTX-M-15}$ among SSIs, HSs and arthropods(Continued)**

| Source | Variables | | Positive (*n*) | Positive (%) | Total (*n*) | Overall significance | Significance (compared to the reference category[a]) | OR | 95% CI for OR | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | Lower | Upper |
| | $bla_{CTX-M-15}$ | | | | | | | | | |
| | Arthropods | Moths[a] | 55 | 28.35 | 194 | | | | | |
| | | Ants | 9 | 8.18 | 110 | | <0.001 | 0.2 | 0.1 | 0.5 |
| | | Spiders | 24 | 36.36 | 66 | | 0.22 | 1.4 | 0.8 | 2.6 |
| | | **Cockroaches** | 236 | 44.78 | 527 | | <0.001 | 2.1 | 1.4 | 2.9 |
| | | Flies | 368 | 34.46 | 1,068 | | 0.10 | 1.3 | 0.9 | 1.9 |
| | Season | Summer[a] | 448 | 36.16 | 1,239 | NA | 0.25 | 0.9 | 0.7 | 1.1 |
| | | **Winter** | 244 | 33.61 | 726 | | | | | |
| | Ward | Male[a] | 184 | 28.93 | 636 | NA | <0.001 | 1.5 | 1.2 | 1.9 |
| | | Female | 508 | 38.22 | 1,329 | | | | | |
| | $bla_{OXA-48}$-like | | | | | | | | | |
| | Arthropods | Moths[a] | 3 | 1.55 | 194 | | | | | |
| | | Ants | 3 | 2.73 | 110 | | 0.48 | 1.8 | 0.4 | 9.0 |
| | | Spiders | 1 | 1.52 | 66 | | 0.99 | 1.0 | 0.1 | 9.6 |
| | | Cockroaches | 31 | 5.88 | 527 | | 0.024 | 4.0 | 1.2 | 13.2 |
| | | Flies | 13 | 1.22 | 1,068 | | 0.71 | 0.8 | 0.2 | 2.8 |
| | Season | Summer[a] | 22 | 1.78 | 1,239 | NA | 0.004 | 2.3 | 1.3 | 4.0 |
| | | Winter | 29 | 3.99 | 726 | | | | | |
| | Ward | Male[a] | 11 | 1.73 | 636 | NA | 0.10 | 1.8 | 0.9 | 3.5 |
| | | Female | 40 | 3.01 | 1,329 | | | | | |
| HS | $bla_{NDM}$, $bla_{CTX-M-15}$ and $bla_{OXA-48}$-like | | | | | | | | | |
| | Season | Summer[a] | 72 | 23.38 | 308 | | | | | |
| | | Winter | 91 | 28.80 | 316 | NA | 0.12 | 1.3 | 0.9 | 1.9 |
| | Ward | Male | 88 | 24.93 | 353 | | | | | |
| | | Female | 75 | 27.68 | 271 | NA | 0.44 | 1.2 | 0.8 | 1.7 |
| | Origin | Door surfaces[a] | 18 | 10.23 | 176 | | | | | |
| | | **Near non-clinical waste** | 58 | 65.91 | 88 | <0.001 | <0.001 | 17.0 | 8.8 | 32.7 |
| | | Bathroom button | 11 | 12.50 | 88 | | 0.58 | 1.3 | 0.6 | 2.8 |
| | | Cupboard handle | 13 | 14.77 | 88 | | 0.28 | 1.5 | 0.7 | 3.3 |
| | | **Window surfaces** | 19 | 43.18 | 44 | | <0.001 | 6.7 | 3.1 | 14.4 |
| | | Bed surfaces | 10 | 19.23 | 52 | | 0.09 | 2.1 | 0.9 | 4.9 |
| | | **Sink knob** | 34 | 38.64 | 88 | | <0.001 | 5.5 | 2.9 | 10.6 |

The results of the logistic regression analysis on the associations of the carriage of target resistance genes (outcome variable) with the explanatory categorical variables of interest within SSIs, arthropods and HSs are shown. The explanatory variables were considered individually with an overall significance for the variable given along with the significance of each category of the variable compared to a selected reference category. The individual *P* values were not corrected for multiple comparisons; however, the directionality and magnitude of the difference between categories and the reference category can be observed by considering the ORs between each category and the reference category. Statistically significant *P* values are indicated in bold. The reference for SSIs was taken as stays of 1–3 d; ward, male; and season, summer. The reference for arthropod variables was taken as insect, moths; ward, male; and season, summer. The reference for HS variables was taken as season, summer. The significant associations are shown separately for each gene among SSIs and arthropods, and collectively among HS samples. $P < 0.05$ was considered to be significant. [a]Reference variables

between MDRE from SSIs and certain HSs, in particular locations near non-clinical waste-bins (OR = 17.0, 95% CI = 8.8–32.7), window surfaces (OR = 6.7, 95% CI = 3.1–14.4) and sinks (OR = 5.5, 95% CI = 2.9–10.6) when compared with door surfaces (10.2%, 18/176) (Table 1, Fig. 2 and Supplementary Table 4). Among arthropods, cockroaches showed a significant association with the carriage of $bla_{NDM}$ (OR = 3.7, 95% CI = 2.2–6.2), $bla_{OXA-48}$-like (OR = 4.0, 95% CI = 1.2–13.2) and $bla_{CTX-M-15}$ (OR = 2.1, 95% CI = 1.4–2.9) when compared to carriage by moths. Furthermore, carriage of $bla_{CTX-M-15}$ (OR = 1.5, 95% CI = 1.2–1.9) among arthropods was associated with being in female wards when compared with being in male wards (Table 1 and Supplementary Table 3).

**Bacterial species and antibiotic-resistance profiling.** To identify the bacterial species that are responsible for the carriage of resistance genes, we performed matrix-assisted laser desorption/ionization–time of flight (MALDI–TOF) mass spectrometry analysis or 16S rRNA sequencing if identification was not achieved through MALDI–TOF analysis of PCR-positive isolates. Totals of 52, 30 and 19 different species were recovered from arthropod, SSI and




| | TGC | FOS | CIP | CN | F | AMC | CTX | CAZ | FEP | IMI | ETP | MRP | ATM | CS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SSIs | 2.33 | 6.98 | 83.72 | 58.14 | 48.84 | 100.0 | 88.37 | 88.37 | 67.44 | 23.26 | 83.72 | 44.19 | 72.09 | 9.30 |
| HSs | 2.44 | 14.63 | 70.73 | 51.22 | 36.59 | 92.68 | 82.93 | 80.49 | 73.17 | 36.59 | 92.68 | 39.02 | 60.98 | 0.00 |
| Arthropods | 2.82 | 22.03 | 66.10 | 55.37 | 27.12 | 94.35 | 96.05 | 93.22 | 83.62 | 7.91 | 78.53 | 16.95 | 64.41 | 1.69 |

**Fig. 3 | Antibiotic resistance profiles among different sources.** The resistance percentage of $bla_{\text{NDM}}$ and $bla_{\text{OXA-48}}$-like PCR-positive samples isolated from SSIs ($n=43$), HSs ($n=40$) and arthropods ($n=177$). AMC, amoxicillin-clavulanic acid; ATM, aztreonam; CAZ, ceftazidime; CIP, ciprofloxacin; CN, gentamycin; CS, colistin; CTX, cefotaxime; F, nitrofurantoin; FEP, cefepime; FOS, fosfomycin; IPM, imipenem; MEM, meropenem; TGC, tigecycline.

HS samples, respectively. From SSIs ($n=177$), we recovered *E. coli* (36.2%), *Enterobacter* spp. (20.3%), *Klebsiella* spp. (18.6%) and *Pseudomonas* spp. (7.9%). From HS isolates ($n=107$), we recovered 34.6% *Klebsiella* spp., 27.1% *Enterobacter* spp., 8.4% *Routella* spp. and 7.5% *Pseudomonas* spp. Bacterial isolates ($n=549$) from arthropods also possessed a high number of Gram-negative bacteria (GNB), including *Enterobacter* spp. (26.0%), *Citrobacter* spp. (23.7%), *E. coli* (15.1%), *Klebsiella* spp. (14.0%) and *Pseudocitrobacter* spp. (8.9%) (Supplementary Fig. 4).

Antimicrobial susceptibility profiles of $bla_{\text{NDM}}$ and $bla_{\text{OXA-48}}$-like PCR-positive isolates ($n=262$) encompassing 13 genera and 27 species showed high levels of resistance to fluoroquinolones and aminoglycosides. Resistance to ciprofloxacin was 83.7% for GNB isolated from SSIs, 70.7% for HSs and 66.1% for arthropods. Similar resistance profiles to gentamicin were observed for GNB isolated from all sources (SSIs, 58.1%; HSs, 51.2%; and arthropods, 55.4%). Ertapenem was shown to be the least active antibiotic amongst all carbapenems with 83.7%, 92.7% and 78.5% resistance for GNB isolated from SSIs, HSs and arthropods, respectively. Resistance to meropenem was 44.2%, 39.0% and 17.0% for GNB isolated from SSIs, HSs and arthropods, respectively. Tigecycline, fosfomycin and colistin were most active among all of the antibiotics tested (Fig. 3 and Extended Data Fig. 2).

**Interspecies diversity, carbapenemase variants and plasmids.** We investigated interspecies diversity and performed WGS on PCR-positive *E. coli* ($n=117$), *E. cloacae* ($n=122$) and *K. pneumoniae* ($n=96$) from SSIs, HSs and arthropods because they were found to be the most dominant MDRE species. WGS analysis revealed populations consisting of diverse sequence types (STs) carrying multiple resistance genes and plasmids with no clear correlations between samples from different sources.

*E. coli* ($n=117$) from this study displayed 38 different STs, and most predominant among these STs were ST410 ($n=23$), ST167 ($n=14$) and ST354 ($n=11$) (Extended Data Fig. 3). ST410 was also found to be the dominant *E. coli* ST group in this study, carrying $bla_{\text{OXA-181}}$ and/or $bla_{\text{CTX-M-15}}$, and could potentially be an emerging dominant global clone[22,23]. Similarly, the established global dominant clone of *E. coli*, ST131, was identified among five *E. coli* isolates from four patients and a spider and carried $bla_{\text{CTX-M-15}}$ in accordance with previously published data[24]. Eight previously unknown *E. coli* O-serotypes were identified corresponding to 11/50 of the observed O:H combinations. Onovel32:H9 was frequently isolated ($n=19$), followed by O45:H6 ($n=11$) and O8:H9 ($n=11$). Onovel32:H9 was mostly associated with ST167 ($n=10/19$), O45:H6 with ST354 ($n=9/11$) and O8:H9 with ST410 ($n=9/11$) (Supplementary Table 6). One hundred samples carried a variant *of* $bla_{\text{CTX-M}}$, mainly $bla_{\text{CTX-M-15}}$ ($n=86$) and, to a lesser extent, $bla_{\text{CTX-M-24}}$ ($n=11$), and $bla_{\text{CTX-M-27}}$, $bla_{\text{CTX-M-55}}$ and $bla_{\text{CTX-M-71}}$ ($n=1$). Twenty-nine samples from SSIs, HSs and arthropods harboured $bla_{\text{OXA-181}}$ spread across ST410 ($n=18$) and ST354 ($n=11$). Twelve *E. coli* from SSIs and arthropods harboured $bla_{\text{NDM}}$ variants ($bla_{\text{NDM-1}}$ ($n=4$), $bla_{\text{NDM-4}}$ ($n=1$), $bla_{\text{NDM-5}}$ ($n=5$) and $bla_{\text{NDM-7}}$ ($n=2$) spread across a diversity of STs (ST101, ST167 and ST410 and ST940 ($n=2$); ST224, ST448, ST2659 and ST5653 ($n=1$)). Twenty-three $bla_{\text{OXA-181}}$ and 10 $bla_{\text{NDM}}$ positive bacteria also harboured $bla_{\text{CTX-M}}$ (Supplementary Table 5).

Among *E. cloacae* ($n=122$), 49 different STs, including 10 new STs, and 6 previously unknown alleles were recovered (Extended Data Fig. 4). The dominant STs were identified as ST114 ($n=9$), ST946 ($n=9$) and ST109 ($n=9$). Eighty-three samples from SSIs, HSs and arthropods harboured $bla_{\text{CTX-M-15}}$, 36 harboured $bla_{\text{NDM}}$ variants ($bla_{\text{NDM-1}}$ ($n=33$) and $bla_{\text{NDM-7}}$ ($n=3$)) and 8 harboured $bla_{\text{OXA-181}}$. Eleven $bla_{\text{NDM-1}}$ and two $bla_{\text{OXA-181}}$ containing isolates coincided with $bla_{\text{CTX-M-15}}$. Six out of eight isolates carrying $bla_{\text{OXA-181}}$ were ST116. Interestingly, among *E. cloacae*, certain STs were all carriers of $bla_{\text{NDM}}$ variants or $bla_{\text{OXA-181}}$. For example, all isolated ST1000 ($n=5$) and ST1458 ($n=5$) carried $bla_{\text{NDM-1}}$, and all ST93 ($n=5$) carried $bla_{\text{NDM-1}}$ or $bla_{\text{NDM-7}}$. Additional prominent ST of $bla_{\text{NDM-1}}$ included ST109 ($n=6$). However, resistance genes were spread across a variety of STs and no particular association was observed (Supplementary Table 5).

Large diversity was observed for *K. pneumoniae* ($n=96$), represented by 54 STs and 41 different KL serotypes (Extended Data Fig. 5). *K. pneumoniae* isolates from our study were dominated by a high-risk clone of ST15 ($n=10$), six of which carried $bla_{\text{NDM-1}}$ and five were KL112. All ST15 isolates from either SSIs or HSs ($n=5$) contained the virulence factor (VF) yersiniabactin (*ybt*), whereas only 1/5 ST15 *K. pneumoniae* isolate cultures from arthropods contained *ybt*. ST15 is mostly seen as a human pathogen but is increasingly identified as a dominant ST from animals[25]. We also identified

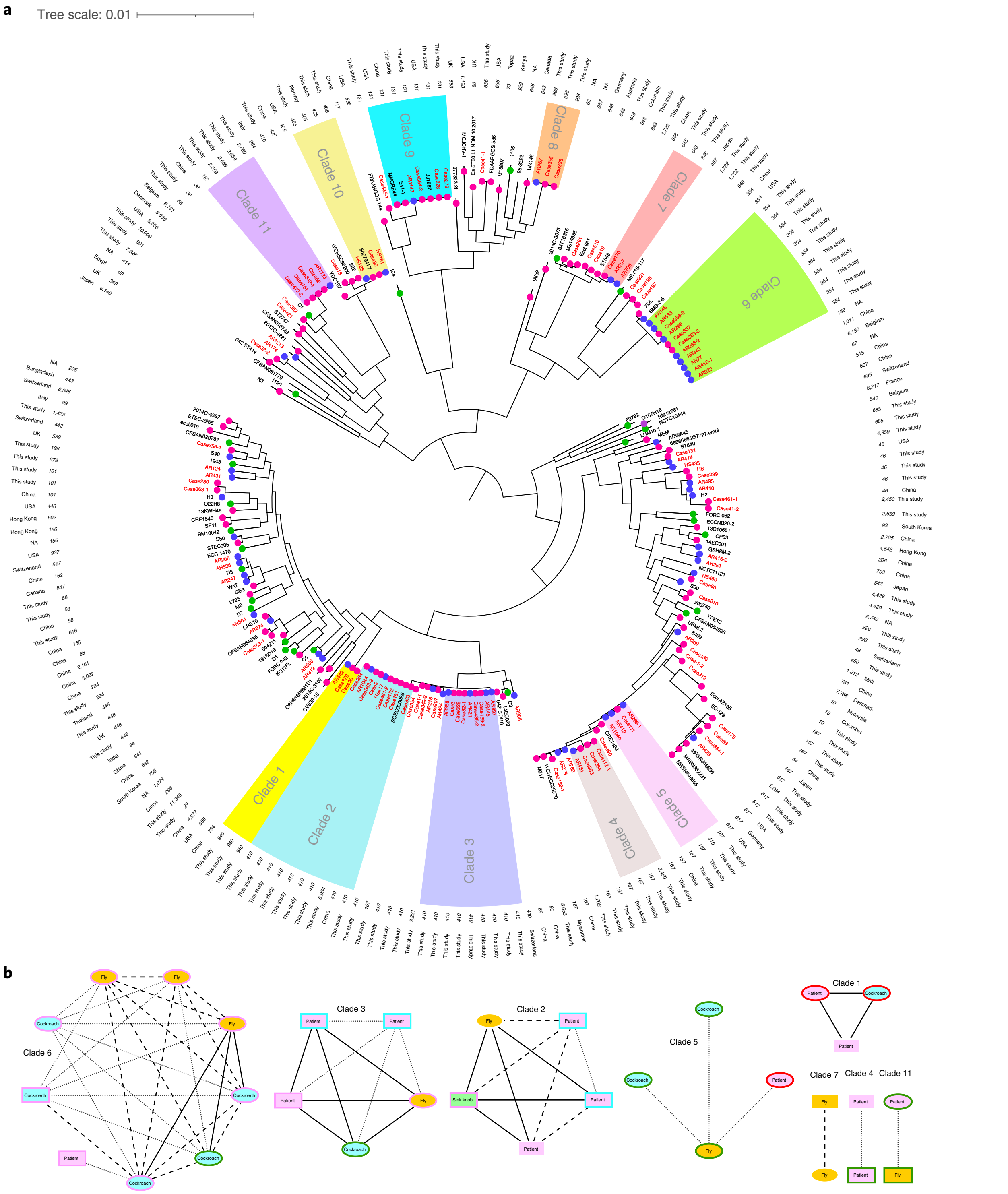



**Fig. 4 | *E. coli* phylogenetic and transmission analysis.** Phylogenetic and transmission analysis of *E. coli*. **a**, The core phylogenetic tree between samples derived from this study, including the reference genomes extracted from NCBI database. The sources are shown as coloured circles on the branches (SSIs from this study and clinical samples from databases (pink); arthropods (AR) from this study and animals originating from databases (blue); HSs from this study and environmental samples from databases (green)). ST and geographical location are displayed as text on the outside. Isolates of this study are labelled in red font, and clades of ≥3 isolates from this study are shown in different coloured ranges. Missing information is denoted by NA (not available). **b**, The Network chart between samples of different origins with ≤20 SNPs is shown. Connections between samples are represented by different styles of lines that correspond to the number of SNPs (0–5 (solid lines), 6–10 (dashed lines) and 11–20 (dotted lines) SNPs). The background colour represents the origin of the isolates (SSIs (pink); HSs (green) and arthropods (blue (cockroaches) and orange (flies)) and the shape represents the ward of isolation (female (round) and male (rectangle)). The presence of target resistance genes for each sample is shown by the outline colour (*bla*$_{\text{CTX-M-15}}$ (green), *bla*$_{\text{CTX-M-15}}$ and *bla*$_{\text{NDM}}$ (black), *bla*$_{\text{CTX-M-15}}$ and *bla*$_{\text{OXA-48}}$-like (pink), *bla*$_{\text{NDM}}$ (red) and *bla*$_{\text{OXA-48}}$-like (blue)). The exact location of surfaces or species of insect is shown as text within the network edges.

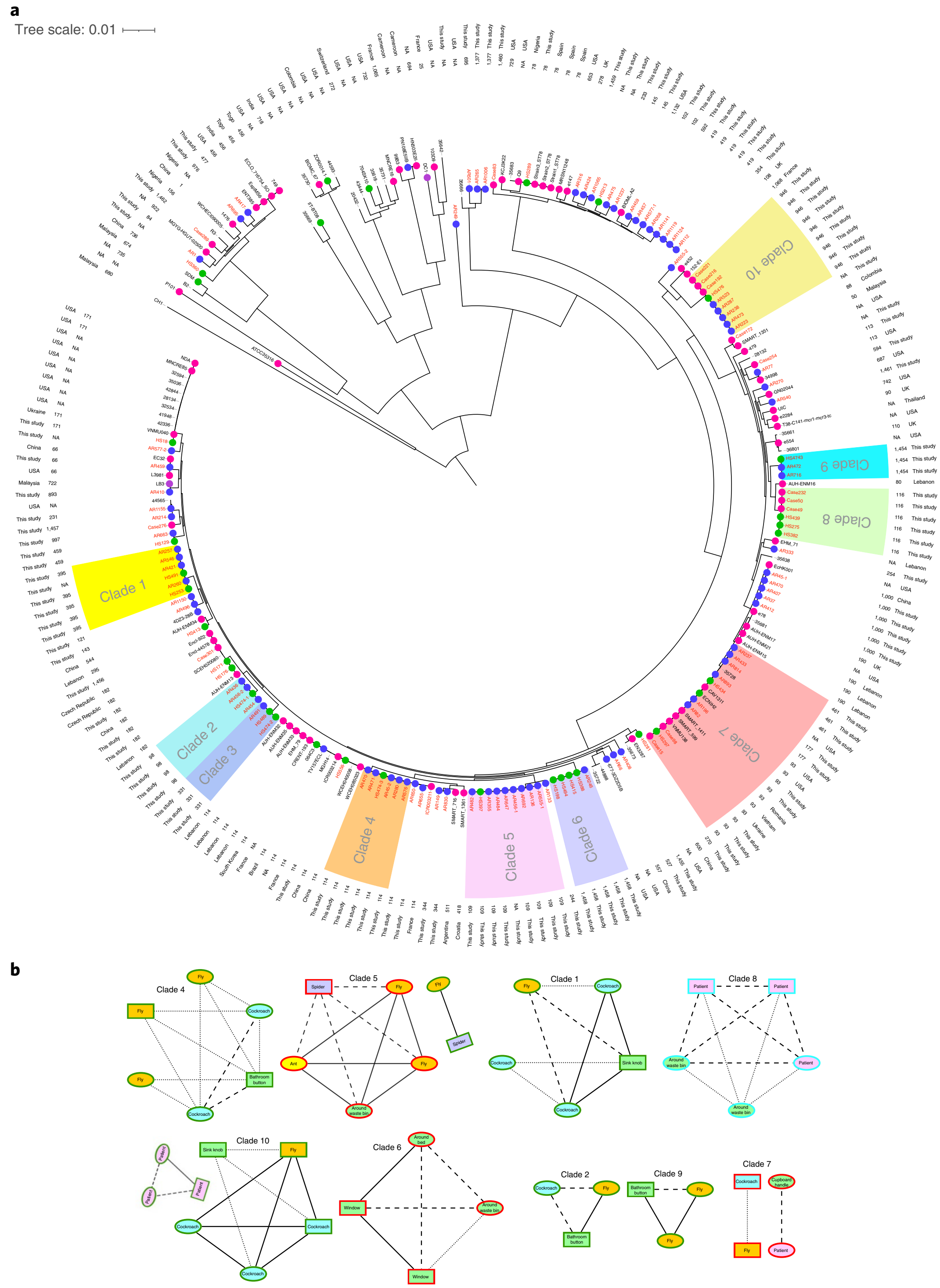



**Fig. 5 | *E. cloacae* phylogenetic and transmission analysis.** Phylogenetic and transmission analysis of *E. cloacae*. **a**, The core phylogenetic tree between samples derived from this study, including the reference genomes extracted from the NCBI database. The sources are shown as coloured circles on branches (SSIs from this study and clinical sample from databases (pink); arthropods from this study and animals originating from databases (blue); HSs from this study and environmental samples from databases (green)). ST and geographical location are displayed as text on the outside. Isolates of this study are labelled in red font and clades of ≥3 isolates from this study are shown in different coloured ranges. Missing information is denoted by NA (not available). **b**, The network chart between samples of different origins with ≤20 SNPs is shown. Connections between samples are represented by different styles of lines corresponding to the number of SNPs (0–5 (solid lines), 6–10 (dashed lines) and 11–20 (dotted lines) SNPs). The background colour represents the origin of the isolates (SSIs (pink); HSs (green) and arthropods (blue (cockroach), yellow (ants), purple (spiders) and orange (flies)) and the shape represents the ward of isolation (female (round) and male (rectangle)). The presence of target resistance genes for each sample is shown by the outline colour (*bla*$_{CTX-M-15}$ (green), *bla*$_{CTX-M-15}$ and *bla*$_{NDM}$ (black), *bla*$_{CTX-M-15}$ and *bla*$_{OXA-48}$-like (pink), *bla*$_{NDM}$ (red) and *bla*$_{OXA-48}$-like (blue)). The exact location of the surfaces or species of insect is shown as text within the network edges.

a number of *K. pneumoniae* ST11 ($n=5$) and ST231 ($n=6$), which are mainly reported from Asian countries[25,26]. In contrast to ST231 and ST15—which showed similar characteristics to other studies, and carried $bla_{OXA-232}$ and $bla_{NDM-1}$, respectively—ST11 from our study predominantly carried $bla_{CTX-M-15}$ and no carbapenemase genes were identified. Twenty-seven isolates harboured $bla_{NDM}$ ($bla_{NDM-1}$ ($n=24$), $bla_{NDM-5}$ ($n=2$) and $bla_{NDM-7}$ ($n=1$)) and eight $bla_{OXA}$ carbapenemases ($bla_{OXA-181}$ ($n=4$) and $bla_{OXA-232}$ ($n=4$)) spanning over 17 and 5 STs, respectively. Seventy-five isolates harboured $bla_{CTX-M-15}$, coinciding with 20 and 5 samples harbouring $bla_{NDM-1}$ and $bla_{OXA-181/232}$, respectively. Multiple different capsule types ($n=51$KL-loci and $n=14$ O-locus) were identified and all ST231 contained KL51/O1v2, three contained *ybt* VF and two of these isolates also contained the VF *iuc* (Supplementary Table 5 and Extended Data Fig. 5).

**Transmission networks for MDRE.** To examine the relatedness between samples, phylogenetic relationships were further investigated by SNP analysis, including the spatiotemporal connectivity between samples of different origins. Regardless of their origin, the majority of genomes from this study clustered into tight phylogenetic clades, indicating that they may represent the spread of local lineages. The only exceptions were globally dominant STs of *E. coli* ST131, *K. pneumoniae* ST15 and ST231, and *E. cloacae* ST93, ST177 and ST461 (Figs. 4 and 5 and Extended Data Fig. 6).

The degree of intraclade relatedness or 'local clusters' between HS, arthropod and SSI samples derived from male and female wards was assessed by pairwise SNP distances. We set the cut-off threshold for clonality at ≤20 SNPs on the basis of the concordance between the sensitivity analyses performed herein (Supplementary Fig. 7) and published mutation rates and outbreak SNP cut-off values. The rate of substitution is estimated to be 1 SNP per genome per year in *E. coli* and ~4 SNPs per genome per for *K. pneumoniae* and *E. cloacae*, whereas the cut-off for outbreak has been suggested to be <10 SNPs[27,28]. Multiple links between samples within and between different sources were found under this threshold, indicating relatively recent transmission events (Figs. 4 and 5, and Extended Data Figs. 6 and 8–10).

For the initial SNP analysis, both isolates from the representative global collection within the same clade in the core genome phylogenetic analysis (Figs. 4–5 and Extended Data Fig. 6), and from the entire NCBI download shortlisted following a large Mash distance genome index analysis were incorporated. All NCBI genomes were later eliminated from the transmission analysis as they were found to be too diverse to the isolates of this study (high number of SNPs), providing further evidence of the spread of local clusters (Supplementary Figs. 5–7).

*E. coli*, *E. cloacae* and *K. pneumoniae* expressed $n=57$, $n=66$ and $n=11$ linkages below the ≤20 SNP threshold, respectively. *E. cloacae* was most commonly distributed among environmental samples and therefore produced the most linkages between arthropods and HS. Twenty-seven intersource linkages were identified, including $n=20$ connections among HSs and arthropods and $n=7$ connections between SSIs and HSs (Fig. 5, Supplementary Fig. 7 and Extended Data Fig. 10). Among $n=39/66$ intrasource linkages, $n=6$, $n=7$ and $n=26$ were found among SSIs, HSs and arthropods, respectively.

Local *K. pneumoniae* isolates were found to be highly diverse, resulting in fewer local clades and fewer linkages within clades. Only $n=3/11$ *K. pneumoniae* clades displayed ≤20 SNPs between samples from different sources—arthropods and HSs ($n=1$), and SSIs and HSs ($n=2$). Among 8/11 intrasource linkages, $n=2$ connections were identified between SSI and $n=6$ between arthropod samples (Supplementary Figs. 6 and Extended Data Fig. 9). By contrast, *E. coli* was found to display more linkages ($n=18/57$) between samples from different origins. The majority of *E. coli* linkages occurred between arthropods and SSIs ($n=14$) and, to a lesser extent, between HSs and SSIs ($n=4$). Among 39/57 intrasource linkages, $n=8$ connections were identified between SSI and $n=31$ between arthropod samples (Fig. 4 and Extended Data Fig. 8). Most of these linked isolates from *E. coli*, *K. pneumoniae* and *E. cloacae* carried one or combinations of multiple resistance genes, including different variants of $bla_{CTX-M}$ ($bla_{CTX-M-15}$ and $bla_{CTX-M-24}$), $bla_{OXA}$ ($bla_{OXA-1}$, $bla_{OXA-10}$ and $bla_{OXA-181}$) and $bla_{NDM}$ ($bla_{NDM-1}$, $bla_{NDM-5}$ and $bla_{NDM-7}$).

## Discussion

This clinical and molecular epidemiology study was performed in a typical 'semi-rural' public hospital in Pakistan, serving a population of 2 million people. D'Souza et al. undertook a similar study and reported MDR bacteria in 3.3% and 86.7% of US and Pakistani (Islamabad) surface samples, respectively[29]. We report similar results; 82.1% surface sites were MDR positive at any timepoint in our study. We also investigated the transmission of MDRE isolates among HSs, SSIs and arthropods and report a role for arthropods as a source of MDRE in this hospital (Figs. 4 and 5, Extended Data Fig. 6 and Supplementary Figs. 5–7).

Although the relationship between arthropods and MDR/XDRE is unsurprising, we report carriage of MDRE bacteria in arthropods and clonal links with human clinical samples. KTH has limited infection control policies and inadequate waste management, with hospital waste routinely discarded in public places at a distance of <1 km from the hospital (Supplementary Fig. 2c). These waste deposits serve as reservoirs of MDRE that enable birds, flies and cockroaches to distribute MDR/XDR isolates (Supplementary Fig. 2). Regrettably, the KTH public hospital that we studied in Pakistan is representative of publicly funded hospitals in South Asia and Africa, and the incidence of AMR is an ongoing challenge for hospitals in LMICs that provide the main healthcare to lower socioeconomic populations[30,31]. These hospitals are under-resourced, lacking both appropriate waste disposal infrastructure and effective hygiene/disinfection procedures[30–32]. The findings of our study can be extrapolated to most healthcare facilities in LMICs, in which arthropods, especially insects, are abundant.

Countries in South Asia report that normal human faecal carriage of $bla_{CTX-M-15}$ is as high as 70% (ref. [33]). Flies feed on waste material, including human and animal faeces; they are therefore likely to acquire $bla_{CTX-M}$-positive bacteria in countries in which sanitation is poor. Notably, population modelling has been applied to predict that the fly population in the UK will increase by approximately 250% by 2080 owing to an increase in global warming (Supplementary Fig. 8), with South Asian countries ranked in the top ten of those countries that are most likely to be affected by climate change[34,35]. Increased flooding as a result of global change, together with unsanitary waste disposal, inadequate sewage infrastructure, reduced infection prevention and control and weak health systems, could provide an ideal habitat for flies and other insects to thrive and disseminate MDRE/XDRE in LMICs and, in light of our study, in hospitals.

The main limitation of our study was the elimination of PCR-negative isolates from genomic and SNP analyses and not including samples from healthcare workers, which would have enabled us to obtain a more robust network. Our study was not designed to capture individualized transmission events or infer causality and, to maximize the capture of clades circulating within the hospital over an extended time period, we applied a broad SNP cut-off.

Our data indicate that arthropods serve as vectors for MDRE and enable their transmission in hospitals, including in patients presenting with SSIs. We investigated region-specific social and demographic factors to understand drivers of resistance carriage and nosocomial infections in LMICs. We recommend that MDR surveillance is informed by One Health AMR policies, including adjusting for seasonal variations and the inclusion of arthropod control in LMIC hospitals to limit the spread of MDRE, reduce SSIs and improve patient outcomes.

## Methods

**Sample collection.** In this study we analysed the impact of arthropods found within the hospital and their dissemination of MDRE on (1) the contamination of HSs and (2) SSIs. We sampled in the summer and winter to analyse seasonal variation. Samples were collected during peak winter (January 2016) and summer (July–August 2016) from a tertiary care hospital (KTH) in Peshawar, Pakistan, and consisted of three different sources of SSIs, HSs and arthropods. Consent for wound swabs was obtained for 342 patients and ethical approval was obtained from the ethical committee of KTH (Supplementary Data 1 and Supplementary Fig 9). Wounds were swabbed with sterile swabs contacting the exudate/puss and preserved in charcoal agar (Liofilchem) at 4 °C. Moreover, data including name, age, gender, place of residence, length of hospital stay and prescribed antibiotic therapy were also collected (Supplementary Data 1).

The insect and spider samples were mostly caught alive and picked with sterile tweezers from floor/walls whereas a fly trap was used to capture flies. One sample constituted single arthropod collected at a timepoint. Spider and insect samples were stored in sterile containers at 4 °C until processed. HS samples were collected at the same time as SSI and arthropod samples and similar sites were sampled each season. No statistical methods were used to predetermine sample sizes but our sample sizes are similar to those reported in previous publications[9–18,29].

**Sample processing and detection of resistance genes using PCR.** All samples were cultured on media to select for GNB; these were chromogenic agar with vancomycin (10 mg $l^{-1}$) (V), vancomycin (10 mg $l^{-1}$) plus cefotaxime (1 mg $l^{-1}$) (VC), and vancomycin (10 mg $l^{-1}$) plus ertapenem (1 mg $l^{-1}$) (VE) (Liofilchem). Swabs were inoculated directly onto chromogenic agar whereas arthropod samples were macerated in water and suspensions used for culture. Cultures were grown overnight at 37 °C and were examined by PCR for the presence of $bla_{NDM}$, $bla_{OXA48}$-like (including $bla_{OXA-181}$ and $bla_{OXA-232}$), $bla_{KPC}$ (growth on VE) and $bla_{CTX-M-15}$ (growth on VC) (Fig. 1). $bla_{CTX-M-15}$ was chosen as an exemplar ESBL due to its reported prevalence in South Asia[33]. A multiplex PCR was designed using PrimerPlex software (Premier Biosoft) to detect $bla_{NDM}$, $bla_{KPC}$ and $bla_{OXA-48}$-like genes. Overnight-grown bacterial colonies were suspended in 200 µl of water and 1 µl was added to the 18 µl of puReTaq Ready-To-Go PCR Beads (Illustra, 27-9557-02) mixed with 1 µl of dye. The loading dye was prepared by mixing 1,000 µl of 20 mg $ml^{-1}$ of Orange G (Sigma-Aldrich) with 200 µl of 0.5% xylene cyanol (Sigma-Aldrich) prepared in 40% sucrose solution (Sigma-Aldrich). Moreover, detection of $bla_{CTX-M-15}$ and 16S rRNA gene amplification for identification was performed using standard PCR using the Extensor Hi-Fidelity PCR Master Mix (AB- 0794/B, Thermo Fisher Scientific). Template DNA was prepared by suspending a 10 µl loop of bacteria (grown overnight at 37 °C) in 200 µl of water. Each reaction mixture contained 12.5 µl ready master-mix, 0.5 µl of each primer, 10.5 µl of molecular-grade water and 1 µl of template. For 16S rRNA, the amplified DNA was extracted using the Qiagen DNA extraction ki (Qiagen), the concentration was measured using a NanoPhotometer (IMPLEN) and the products with concentrations of higher than 5 ng $\mu l^{-1}$ were sent to Eurofins MWG Operon (Eurofins) for sequencing. Details of the primers and PCR conditions are provided in Supplementary Table 6.

**Identification of bacterial species by MALDI–TOF and MIC.** Positive PCR samples were isolated and characterized to the species level using MALDI–TOF mass spectrometry (Bruker) and/or 16S rRNA sequencing. Antibiotic resistance profiles were determined by antibiotic discs (Oxoid) and/or broth microdilutions where appropriate following the guidelines of the European Committee on Antimicrobial Susceptibility Testing (EUCAST) and the results were interpreted using EUCAST breakpoints (v.10.0, 2020). Enterobacterales breakpoints were used when no breakpoints were advised in the guidelines or PK-PD.

**WGS sequencing and bioinformatics analysis.** WGS was performed on *E. coli*, *K. pneumoniae* and *E. cloacae* using the Miseq (Illumina) system. Genomic DNA was extracted using the QIAmp DNA mini kit (Qiagen) with an additional RNase step, on the automated QIAcube platform (Qiagen). The DNA was quantified using the Qubit fluorometer 3.0 using the dsDNA HS assay kit (Thermo Fisher Scientific). DNA libraries were prepared using Nextera XT V2 (Illumina), with bead-based normalization, according to the manufacturer's guidelines. Bioinformatics analysis was performed using a high-performance computing cluster at Cardiff University (ARRCA) and CLIMB[36]. Trimgalore (v.0.4.3)[37] was used to remove the Nextera adapter sequences and low-quality bases. Reports before and after read trimming were generated using fastqc (v.0.11.2)[38] and collated using MultiQC (v.1.7)[39]. The mean read length and number of sequences provided on the MultiQC reports was used to determine sequencing coverage. Paired-end reads were overlapped using Flash (v.1.2.11)[40] and assembled into contigs using SPAdes (v.3.9.0)[41]. The resulting contigs were mapped back to the fastq using BWA (v.0.7.15)[42] and Samtools (v.1.3.1)[43]. Assembly error correction was performed using Pilon (v.1.22)[44] and assembly metrics were generated using quast (v.2.1)[45]. Antibiotic resistance genes, plasmid incompatibility types and MLSTs were identified using the Centre for Genomic Epidemiology platform and associated databases[46]. VFDB[47] within ABRicate[48] was used to screen contigs for virulence-associated genes. The VF score for *E. cloacae* was a sum of all VF hits >98% coverage and identity, whereas, for *E. coli*, the VF score was determined according to Kaper et al.[49] as a guide. Kleborate[50] was used to determine capsular types for *K. pneumoniae*. ABRicate was also used to detect the in silico O:H serogroups for *E. coli* using the EcOH database[51]. ClermonTyping[52] was used to detect the in silico phylogroup of *E. coli* isolates. Genomes were annotated using Prokka (v.1.12)[53]. Species-wide relatedness analysis was performed using Roary (v.3.12.0)[54] to create a core genome alignment and FastTree (v.2.11.11)[55] was used to generate maximum likelihood phylogenetic trees[54]. Phylogenetic trees were mid-point rooted, visualized and annotated using iTOL[56]. In total, 18,761 *E. coli*, 8,663 *K. pneumoniae* and 1,960 *E. cloacae* were downloaded from the NCBI on 5 March 2020. Mash (v.2.2)[57] distance was used to assess the inclusion of NCBI genomes for *Enterobacter*, *E. coli* and *K. pneumoniae* on the basis of the number of shared hashes against the diversity of shared hashes within each clade. In brief, Mash sketches were generated for all *Enterobacter*, *E. coli* and *K. pneumoniae* NCBI genomes. Similarly, Mash sketches for genomes within each clade of interest in this study were generated. Distance estimations within each clade were used to guide the cut-off level for shared hashes for the genome estimation between the NCBI genomes to genomes within this study. Any NCBI genome with shared hash scores of greater than or equal to the lowest score from within the clade estimations was included in downstream SNP-based phylogenetic analyses, which was performed separately for each clade using snpiphy[58], an automated pipeline that uses snippy–gubbins–raxml mapping to a reference genome. To maximize the accuracy of SNP calling across the whole genome, a local reference was selected within each clade for mapping. The final comparison was made by repeating SNP analysis after excluding samples with increased variances. Network charts were created using Cytoscape, showing samples with pairwise SNP distances of ≤20 on the basis of published mutation rates of 1 SNP per genome per year for *E. coli*, ~4 SNPs per genome per year for *K. pneumoniae* and *E. cloacae*, and <10 SNPs for outbreaks[27,28,59].

**Statistical analysis.** Statistical analysis was performed using SPSS statistics (v.26, IBM). Logistic regression was used to perform univariable analysis of the presence or not of the resistance genes and its association with variables such as age, seasons, duration of hospital stay and ward among SSIs; season, ward and swabbed surface sites among HSs; and season, ward and species of different arthropods among arthropods. No test that required data normality was conducted or required. Each of the variables was treated only as categorical in the logistic regressions used. Categorical variables included ward, season, species of arthropods and surface sites. Continuous variables among patients such as age and duration of hospital stay were grouped. Variables with statistically significant associations with the resistance genes were selected for multivariate analysis again by binary logistic regression. Some HS sites displayed a low resistance rate to each of the individual target genes ($bla_{NDM}$, $bla_{OXA-48}$-like and $bla_{CTX-M-15}$); thus, all resistance genes were grouped to evaluated association with surface sites. $P \leq 0.05$ was considered to be significant and estimated ORs are presented with 95% CIs.

**Reporting Summary.** Further information on research design is available in the Nature Research Reporting Summary linked to this article.

## Data availability

Sequences reads have been submitted to the European Nucleotide Archive (ENA) under the project number PRJEB40861. A list of individual accession numbers is provided in Supplementary Data 2. The following databases were used: NCBI (https://github.com/tseemann/abricate/tree/master/db/ncbi); Resfinder v.4.0 (https://cge.cbs.dtu.dk/services/ResFinder/); Plasmidfinder v.2.0 (https://cge.cbs.dtu.dk/services/PlasmidFinder/); MLST v.2.0.4 (https://cge.cbs.dtu.dk/services/MLST/), with MLST allele and profile data from https://pubmlst.org; Serotype finder (https://bitbucket.org/genomicepidemiology/serotypefinder/src/master). Source data are provided with this paper.

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## Author contributions

B.H. and T.R.W. designed and guided the study and analysis. M.I. and Asadullah Khan provided the epidemiological and clinical dataset and collected the samples in this study. B.H., K.S. and T.R.W. wrote the manuscript. B.H., G.-I.S., L.C., M.M.E.-B., G.L. and Afifah Khan performed microbiology culture and sample processing in Cardiff University. B.H., K.S. and E.P. performed WGS experiments. K.S. and B.H. performed bioinformatics analysis, following guidance from J.P. W.J.W. and B.H. performed statistical analysis.

## Competing interests

The authors declare no competing interests.

## Additional information

**Extended data** is available for this paper at https://doi.org/10.1038/s41564-021-00965-1.

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**Correspondence and requests for materials** should be addressed to Brekhna Hassan.

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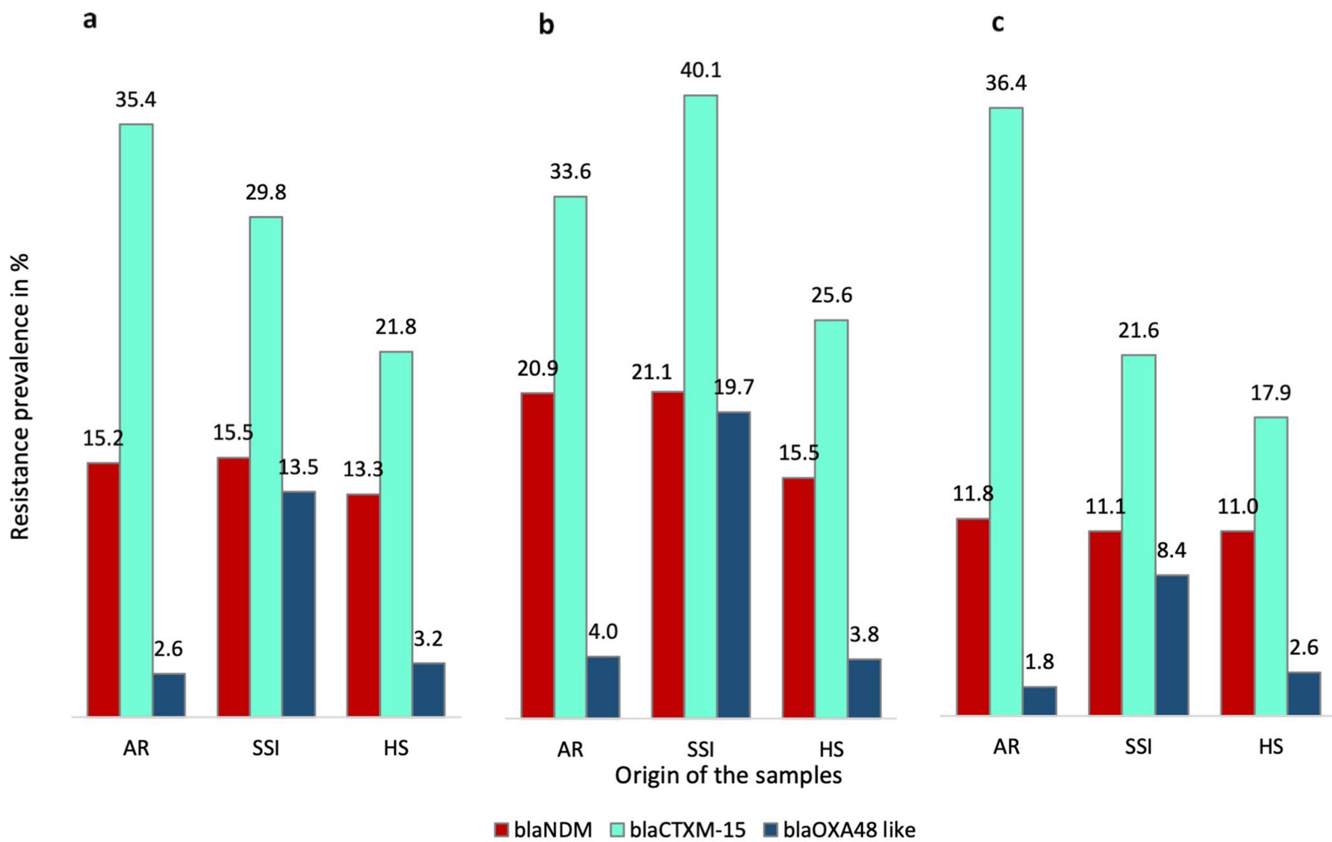
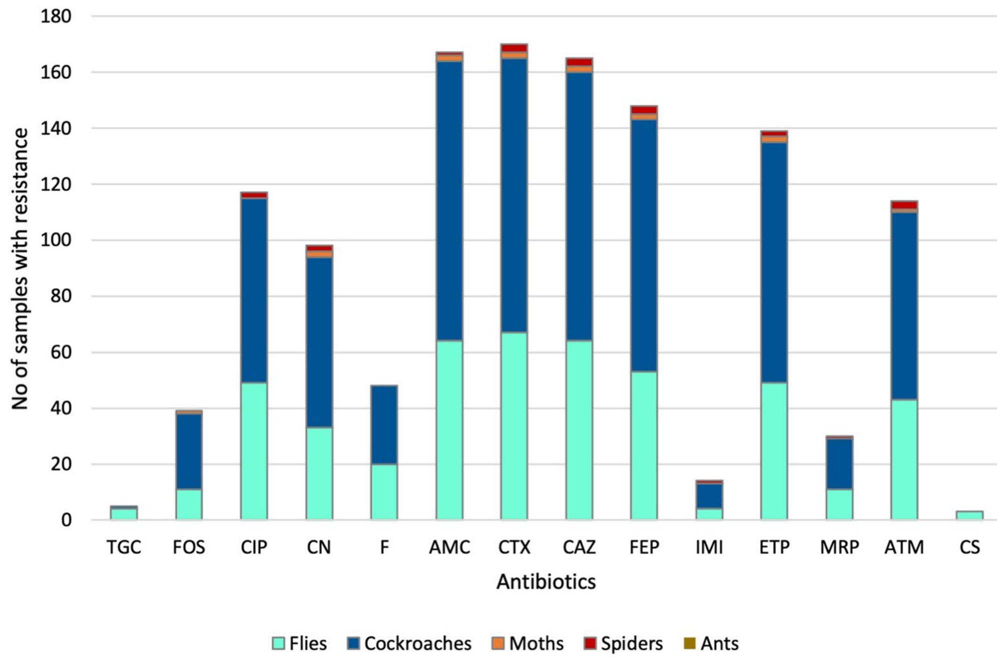
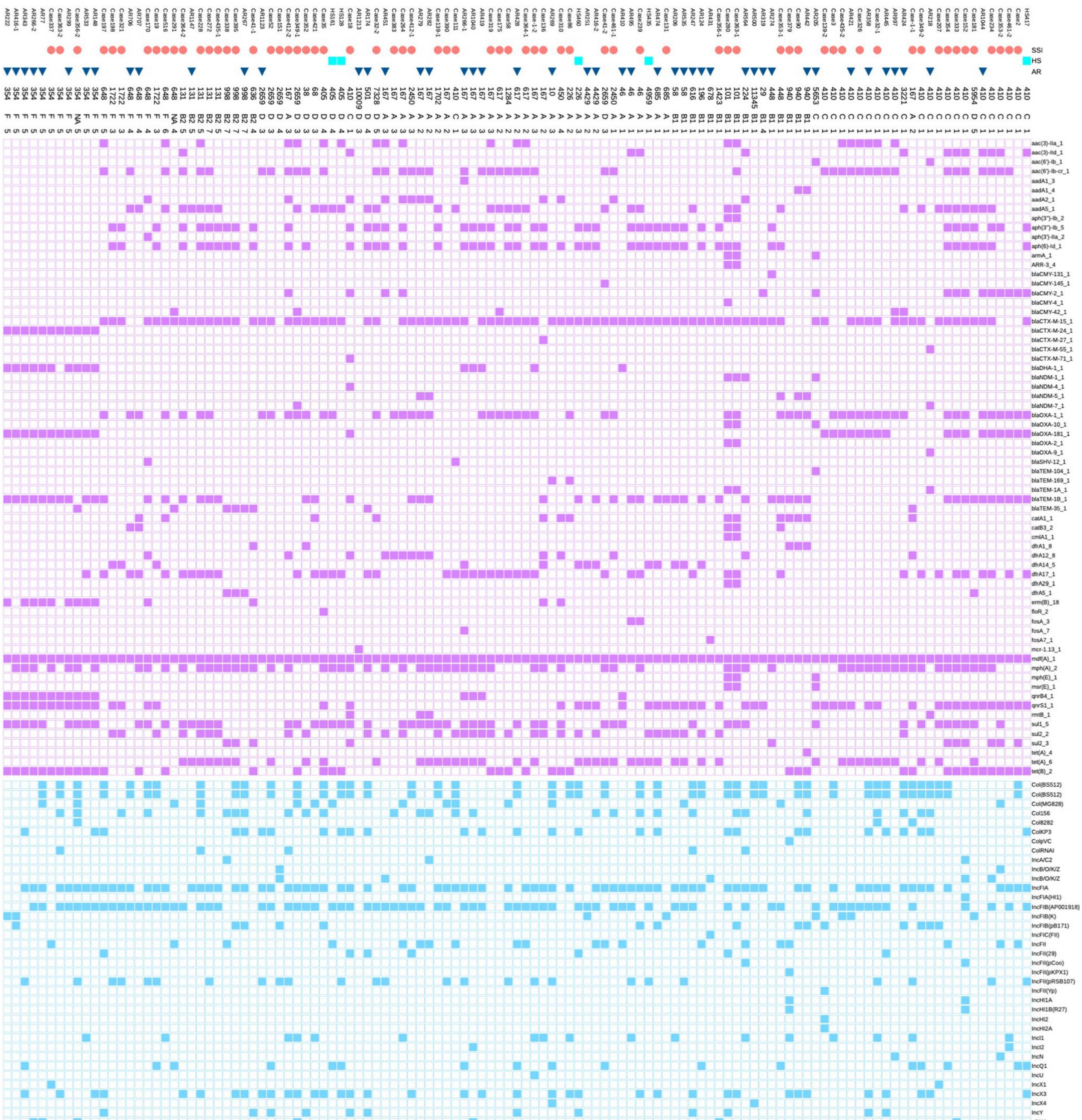



**Extended Data Fig. 1 | Prevalence of resistance genes from different sources and seasons.** Bar chart representing occurrence of blaNDM, blaOXA-48-like and blaCTX-M-15 among collected samples in percentage from arthropods (AR), SSI (surgical site infections) and hospital surfaces (HS) **a**. collective resistance prevalence; **b**. winter; **c**. summer.



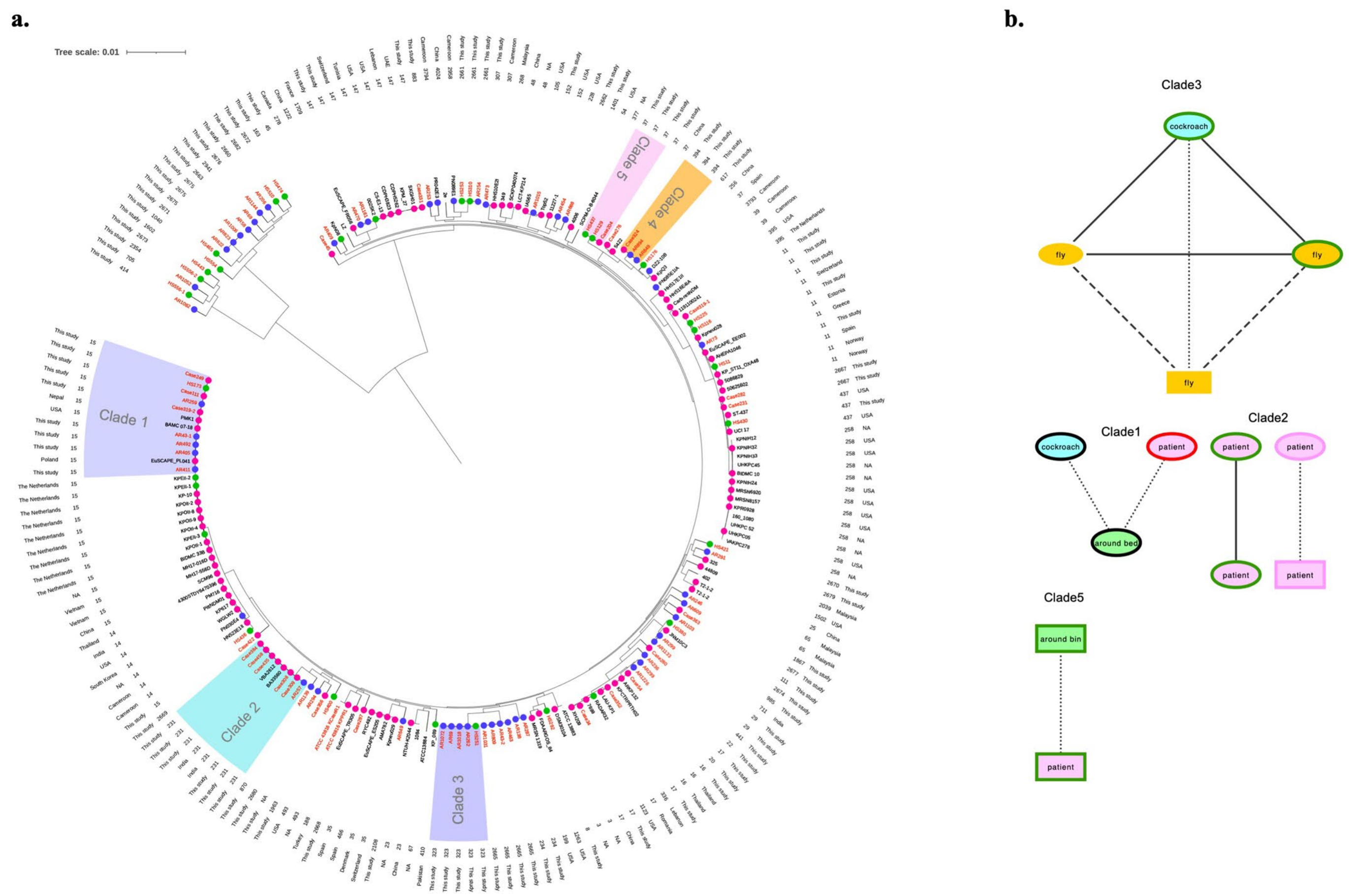

**Extended Data Fig. 2 | Antibiotic resistance profiles of arthropods.** Antibiotic resistance profiles of the $bla_{NDM}$ and $bla_{OXA-48}$-like positive isolates (n = 177) among different species of arthropods. Antibiotics are denoted as TGC = tigecycline; FOS = Fosfomycin; CIP = ciprofloxacin; CN = gentamycin; F = nitrofurantoin; AMC = amoxicillin-clavulanic acid; CTX = cefotaxime; CAZ = ceftazidime; FEP = cefepime; IPM = imipenem; MEM = meropenem; ATM = aztreonam; CS = colistin.



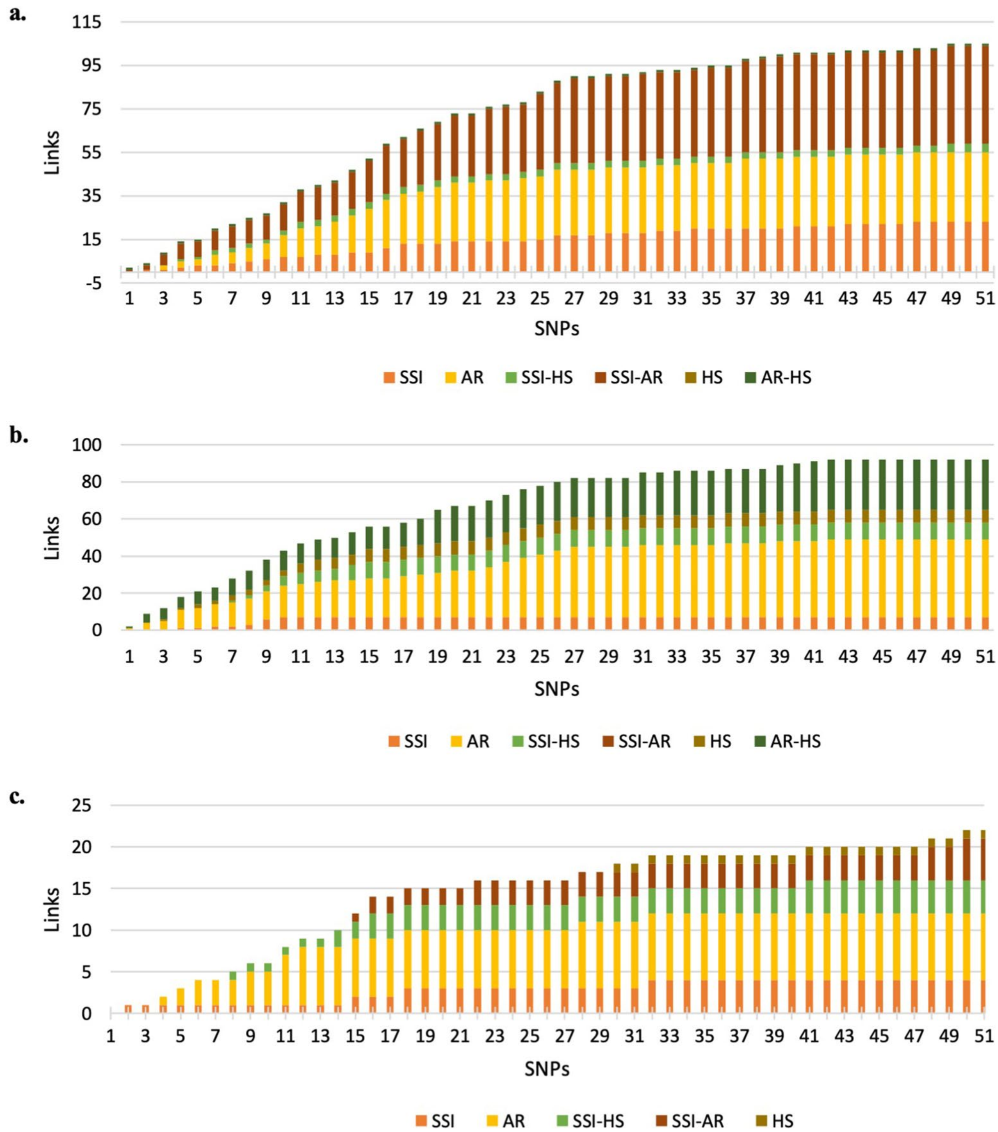
**Extended Data Fig. 3 | *E. coli* WGS results.** Figure showing samples source, STs, phylogroups, virulence score, antibiotic resistance genes and plasmid types among *E. coli* isolated from SSI, HS and arthropods (AR) samples. Sample source is identified by labelled symbols followed by STs, phylogroups and virulence score in text. Presence of antibiotic resistance genes are shown in pink and plasmid Inc groups in blue coloured squares.

**Extended Data Fig. 4 | *E. cloacae* WGS results.** Figure showing samples source, STs, virulence score, antibiotic resistance genes and plasmid types among *E. cloacae* isolated from SSI, HS and arthropods (AR) samples. Sample source is identified by labelled symbols followed by STs and virulence score in text. Presence of antibiotic resistance genes are shown in pink and plasmid Inc groups in blue coloured squares.

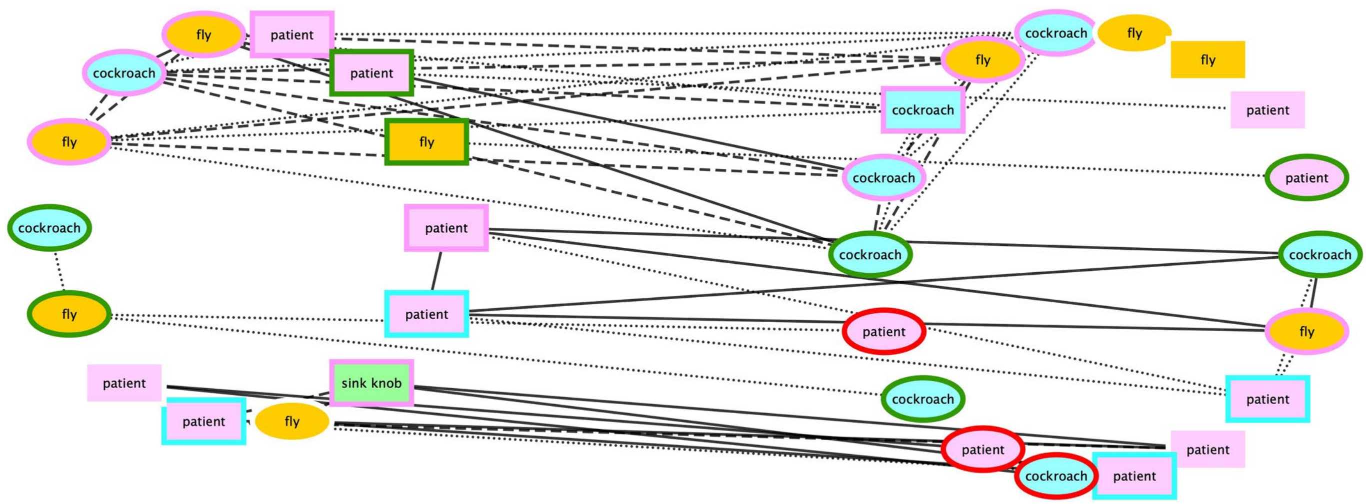
**Extended Data Fig. 5 | *K. pneumoniae* WGS results.** Figure showing samples source, STs, capsular types, virulence score, antibiotic resistance genes and plasmid types among *K. pneumoniae* isolated from SSI, HS and arthropods (AR) samples. Sample source is identified by labelled symbols followed by STs, capsular types and virulence score in text. Presence of antibiotic resistance genes are shown in pink and plasmid Inc groups in blue coloured squares.



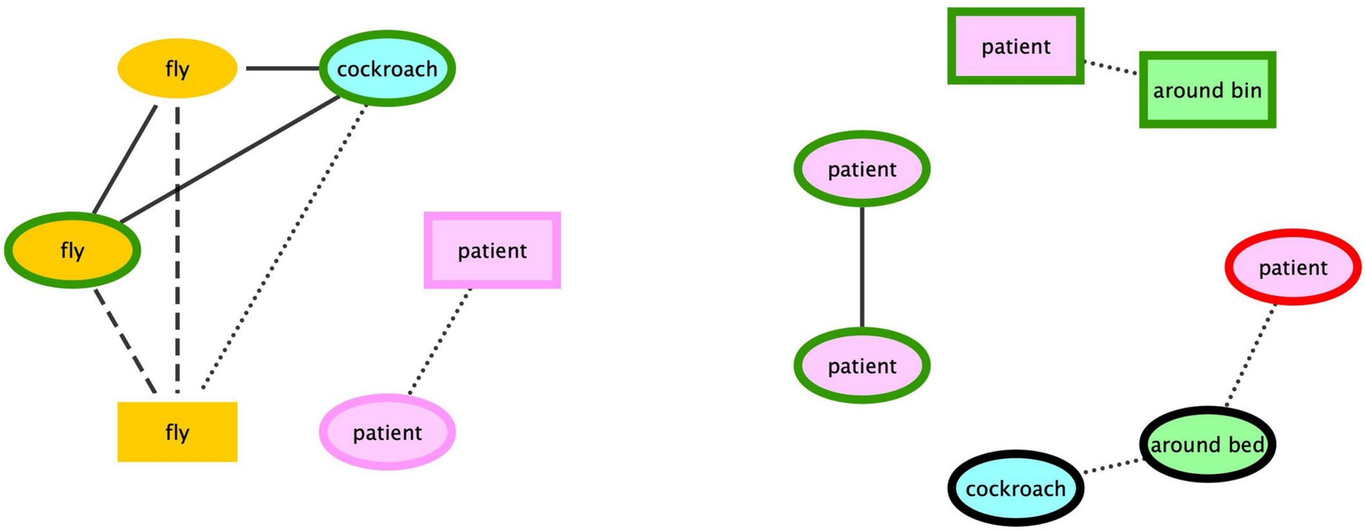

**Extended Data Fig. 6 | *K. pneumoniae phylogenetic* and transmission analysis.** Phylogenetic and transmission analysis of *K. pneumoniae*. **a**, The core phylogenetic tree between samples derived from this study, including the reference genomes extracted from the NCBI database. The sources are shown as coloured circles on branches (SSIs from this study and clinical sample from databases (pink); arthropods from this study and animals originating from databases (blue); HSs from this study and environmental samples from databases (green)). ST and geographical location are displayed as text on the outside. Isolates of this study are labelled in red font and clades of ≥3 isolates from this study are shown in different coloured ranges. Missing information is denoted by NA (not available). **b**, The network chart between samples of different origins with ≤20 SNPs is shown. Connections between samples are represented by different styles of lines corresponding to the number of SNPs (0–5 (solid lines), 6–10 (dashed lines) and 11–20 (dotted lines) SNPs). The background colour represents the origin of the isolates (SSIs (pink); HSs (green) and arthropods (blue (cockroach) and orange (flies)) and the shape represents the ward of isolation (female (round) and male (rectangle)). The presence of target resistance genes for each sample is shown by the outline colour (*bla*$_{CTX-M-15}$ (green), *bla*$_{CTX-M-15}$ and *bla*$_{NDM}$ (black), *bla*$_{CTX-M-15}$ and *bla*$_{OXA-48}$-like (pink), *bla*$_{NDM}$ (red) and *bla*$_{OXA-48}$-like (blue)). The exact location of the surfaces or species of insect is shown as text within the network edges.



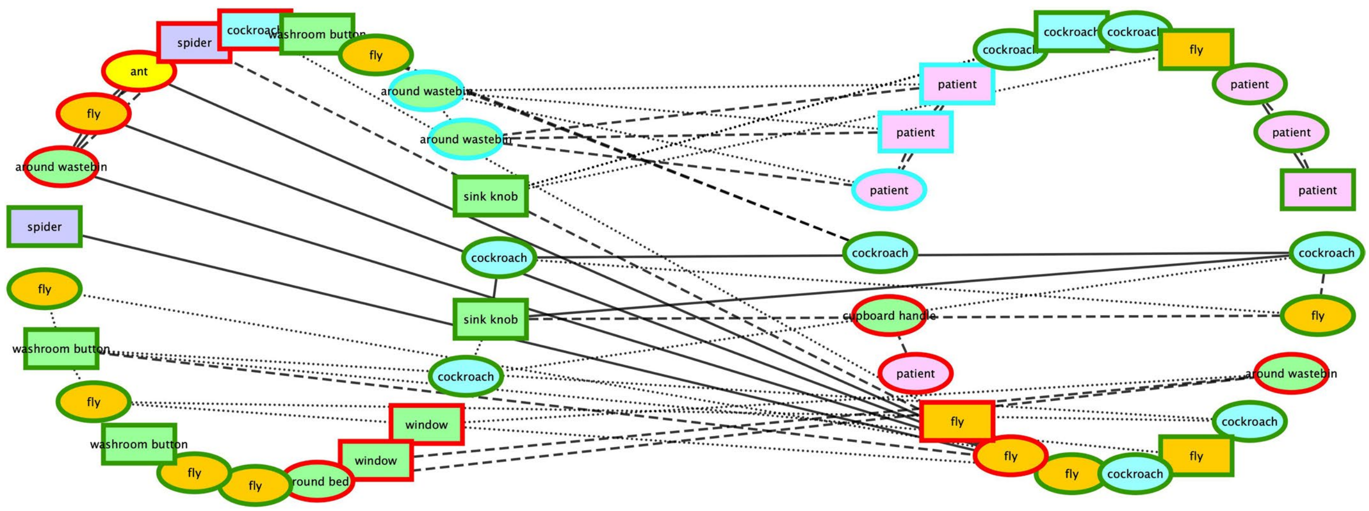

**Extended Data Fig. 7 | SNP threshold sensitivity.** The histogram represents the number of transmission links at each SNP-cutoff point (0-50) among samples between SSI, HS and arthropods (AR) from isolates of **a**. *E. coli*, **b**. *E. cloacae* and **c**. *K. pneumoniae*.




**Extended Data Fig. 8 | Network chart of *E. coli* grouped by seasons.** The network chart between samples collected in summer and winter with ≤20 SNPs. Connections between samples are represented by different styles of lines corresponding to the number of SNPs (—— = 0 - 5, - ● - ● - = 6 - 10 and ● ●●●●●●●● = 11– 20 SNPs). Colours of the edge represent the origin of the isolates (SSI: pink; HS: green and arthropods: blue and orange [cockroach: blue; ant: yellow; spider: purple and fly: orange] and shape represent ward of isolation (round: female and rectangle: male). The presence of target resistance genes for each sample are shown by outline colour of the edges (green: $bla_{CTX\text{-}M\text{-}15}$, black: $bla_{CTX\text{-}M\text{-}15}$ and $bla_{NDM}$, pink: $bla_{CTX\text{-}M\text{-}15}$ and $bla_{OXA\text{-}48}$-like, red: $bla_{NDM}$, blue: $bla_{OXA\text{-}48}$-like). The exact location of surfaces or species of insect is shown as text within the network edges.




**Extended Data Fig. 9 | Network chart of *K. pneumoniae* grouped by seasons.** The network chart between samples collected in summer and winter with ≤20 SNPs. Connections between samples are represented by different styles of lines corresponding to the number of SNPs (—— = 0 – 5, - ● - ● - - = 6 – 10 and ● ●●●●●●●● = 11– 20 SNPs). Colours of the edge represent the origin of the isolates (SSI: pink; HS: green and arthropods: blue and orange [cockroach: blue; ant: yellow; spider: purple and fly: orange] and shape represent ward of isolation (round: female and rectangle: male). The presence of target resistance genes for each sample are shown by outline colour of the edges (green: $bla_{CTX\text{-}M\text{-}15}$, black: $bla_{CTX\text{-}M\text{-}15}$ and $bla_{NDM}$, pink: $bla_{CTX\text{-}M\text{-}15}$ and $bla_{OXA\text{-}48}$-like, red: $bla_{NDM}$, blue: $bla_{OXA\text{-}48}$-like). The exact location of surfaces or species of insect is shown as text within the network edges.




**Extended Data Fig. 10 | Network chart of *E. cloacae* grouped by seasons.** The network chart between samples collected in summer and winter with ≤20 SNPs. Connections between samples are represented by different styles of lines corresponding to the number of SNPs (—— = 0 – 5, **- ● - ● -** = 6 – 10 and ● ●●●●●●●● = 11– 20 SNPs). Colours of the edge represent the origin of the isolates (SSI: pink; HS: green and arthropods: blue and orange [cockroach: blue; ant: yellow; spider: purple and fly: orange] and shape represent ward of isolation (round: female and rectangle: male). The presence of target resistance genes for each sample are shown by outline colour of the edges (green: $bla_{\text{CTX-M-15}}$, black: $bla_{\text{CTX-M-15}}$ and $bla_{\text{NDM}}$, pink: $bla_{\text{CTX-M-15}}$ and $bla_{\text{OXA-48}}$-like, red: $bla_{\text{NDM}}$, blue: $bla_{\text{OXA-48}}$-like). The exact location of surfaces or species of insect is shown as text within the network edges.

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|---|---|---|
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| ☒ | ☐ | A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
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| ☒ | ☐ | A description of all covariates tested |
| ☒ | ☐ | A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
| ☐ | ☒ | A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☐ | ☒ | For null hypothesis testing, the test statistic (e.g. $F$, $t$, $r$) with confidence intervals, effect sizes, degrees of freedom and $P$ value noted<br>*Give P values as exact values whenever suitable.* |
| ☒ | ☐ | For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ | For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☒ | ☐ | Estimates of effect sizes (e.g. Cohen's $d$, Pearson's $r$), indicating how they were calculated |

*Our web collection on statistics for biologists contains articles on many of the points above.*

## Software and code

Policy information about availability of computer code

| | |
|---|---|
| Data collection | Data and questionnaire were collected by doctors working on paper and mailed to us where it was incorporated onto excel sheets. |
| Data analysis | CLIMB (v1.0)<br>Trimgalore (v0.4.3)<br>fastqc (v0.11.2)<br>MultiQC (v1.7)<br>Flash (v1.2.11)<br>SPAdes (v3.9.0)<br>BWA (v.0.7.15)<br>samtools (v1.3.1)<br>Pilon (v1.22)<br>quast (v.2.1)<br>Blast nt (https://blast.ncbi.nlm.nih.gov/Blast.cgi) (v2.2.25)<br>PathogenWatch (v.3.13.10; https://pathogen.watch)<br>srst2 (v0.2.0)<br>ABRicate (v0.9.7)<br>BIGSbd (v1.25.1)<br>Kaptive (v0.7.0)<br>Kleborate (v0.2.0)<br>SerotypeFinder (v2.0)<br>Prokka (v1.12)<br>Roary (v3.12.0)<br>FastTree (v2.1.11) |

iTOL (v5.7)
Mash (v.2.2)
snpiphy (v0.5)
tidyverse package ggplot2 in R (v4.0.2)
Geneious prime (2019.2.3)
Cytoscape (v.3.8.2)
IBM SPSS statistics (v.26)
Excel (v.16.16.27)
Visual Paradigm (v.16.3)
PrimerPlex (v.2.0)

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## Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

Sequences reads have been submitted to the European Nucleotide Archive (ENA) under the project number PRJEB40861. Individual accession numbers are listed in the Source Data under Accession_Codes.xlsx. The following databases were used:

NCBI: https://github.com/tseemann/abricate/tree/master/db/ncbi
Resfinder v4.0: https://cge.cbs.dtu.dk/services/ResFinder/
Plasmidfinder v2.0 https://cge.cbs.dtu.dk/services/PlasmidFinder/
MLST v2.0.4 https://cge.cbs.dtu.dk/services/MLST/, with MLST allele and profile data from https://pubmlst.org
Serotype finder: https://bitbucket.org/genomicepidemiology/serotypefinder/src/masterResfinder v4.0: https://cge.cbs.dtu.dk/services/ResFinder/

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# Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

| | |
|---|---|
| Sample size | The sampling was conducted over a set time-period of winter and summer months to incorporate seasonal variations regardless of their number. This was a time dependent study and not a denominator dependent study; therefore, we simply collected all samples we could during the study period. No statistical methods were used to pre-determine sample sizes but our sample sizes are similar to those reported in previous publications (ref 9-18, 29). |
| Data exclusions | Exclusion criteria were not pre-established. Human samples (n=7) with missing basic information such as gender was excluded from analysis. Arthropods and hospital surface sites samples were only collected from Male and Female wards and hence for SNP analysis patient post-surgical site infection samples originating from wards other than Male and Female were excluded. |
| Replication | Findings were not replicated. The study was designed to identify the prevalence of selected resistance genes and transfer or transmission of bacterial strains and resistance genes from one source to another in hospital settings. Confirmation of results was achieved by whole genome sequencing of selected bacterial species. |
| Randomization | Randomization was not necessary for this study as we were establishing spatiotemporal analysis and patients details along with the locations of the ward, collection point of arthropods and location of the hospital surfaces were essential to establish connections between resistance spread across different sources. |
| Blinding | Blinding was not necessary for this study as we were establishing spatiotemporal analysis and patients details along with the locations of the ward, collection point of arthropods and location of the hospital touch surfaces were essential to establish connections between resistance spread across different sources. |

# Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

## Materials & experimental systems

| n/a | Involved in the study | |
|---|---|---|
| ☒ | ☐ | Antibodies |
| ☒ | ☐ | Eukaryotic cell lines |
| ☒ | ☐ | Palaeontology and archaeology |
| ☐ | ☒ | Animals and other organisms |
| ☐ | ☒ | Human research participants |
| ☒ | ☐ | Clinical data |
| ☒ | ☐ | Dual use research of concern |

## Methods

| n/a | Involved in the study | |
|---|---|---|
| ☒ | ☐ | ChIP-seq |
| ☒ | ☐ | Flow cytometry |
| ☒ | ☐ | MRI-based neuroimaging |

# Animals and other organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research

| | |
|---|---|
| Laboratory animals | The study did not involve laboratory animals |
| Wild animals | The study involved collecting arthropods found in the hospital. Most of the insects and spiders were caught alive whereas a fly trap was employed to capture the flies. Insects captured at the time of sampling were identified as: Musca domestica (house flies), Blattella orientalis or Blattella germanica (cockroaches), Lasius niger (ants) and Aglossa aglossalis (moths). Spiders were identified as Stegodyphus pacificus. All samples were collected in sterile eppendorfs and stored at 4C until processed. |
| Field-collected samples | The study did not involve sample collected from the field. |
| Ethics oversight | The ethical committee of the hospital "Khyber Teaching Hospital" approved the study, signed letter from Dr Mansoor khan dated 11th November 2015. |

Note that full information on the approval of the study protocol must also be provided in the manuscript.

# Human research participants

Policy information about studies involving human research participants

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| Population characteristics | We sampled in summer and winter to analyse seasonal variation. Samples were collected during peak winter (Jan-2016) and summer (July-August 2016) from patients with post-surgical site infections at tertiary care hospital (Khyber Teaching Hospital [KTH]) in Peshawar, Pakistan. The samples were mainly from Male (n= 125, 37.3%) and Female (n=155, 46.3% ) surgical wards with n= 55 (16.4%) from pediatric ward. The initial resistance prevalence was drawn from all samples however, for the SNP transmission analysis the samples collected from pediatric ward with age below 13 years old (local age limit for pediatric ward) were excluded. |
| Recruitment | Patient with post-surgical site infection were recruited with consent for the study. Upon suspected diagnosis of post-surgical site infections patients were approached by clinical staff for enrollment to the study and a swab of the infected area was taken following consent. We adhered to international criteria for SSIs and the expertise of the clinical staff at the time to diagnose a SSI which is not difficult to do so therefore the bias is likely to be minimal at best. |
| Ethics oversight | The ethical committee of the hospital "Khyber Teaching Hospital" approved the study, signed letter from Dr Mansoor khan dated 11th November 2015. Informed consent was obtained with no compensation for the participants. |

Note that full information on the approval of the study protocol must also be provided in the manuscript.